Phase-Primary Quantum Theory (PPQT): Formal Construction

Paper 27 in the Phase Theory Series

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Phase Theory Series Note:

This paper is Paper 27 in a series deriving from Phase Theory (Hanks, 2026 [1]), which proposes that phase is the sole fundamental constituent of physical reality. Readers are assumed familiar with Papers 1–26 of the series. Key results from Papers 3, 5, 6, 9, 14, 19, and 22 are cited but not re-derived in full.

Abstract

We construct Phase-Primary Quantum Theory (PPQT) as the effective quantum description emerging from Phase Theory when admissible phase configurations are projected onto reduced descriptive spaces. PPQT reproduces all operational content of quantum mechanics — superposition, interference, entanglement, operator algebra, and probabilistic outcomes — without invoking wavefunctions, Hilbert spaces, operators as primitives, or measurement postulates. Quantum structure is shown to arise from topology, admissibility basin geometry, and distinguishability measures on phase configuration space. PPQT is therefore not an interpretation of quantum mechanics, but its derivation from phase-first principles.

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1. Introduction and Motivations

Quantum mechanics is among the most empirically successful physical theories ever constructed. Its predictions span phenomena from atomic spectroscopy to quantum computing, from superconductivity to the structure of the cosmos, and in every tested domain it performs with extraordinary precision. Yet for a century, physicists and philosophers have struggled to answer a question that ought to be elementary: *what does quantum mechanics describe?* The operational success of the theory and its foundational intelligibility have remained stubbornly decoupled. This paper is a formal response to that decoupling.

The standard formalism of quantum mechanics rests on six postulates: (i) the state space of a physical system is a complex Hilbert space $\mathscr{H}$; (ii) observables correspond to self-adjoint operators on $\mathscr{H}$; (iii) measurement of an observable yields one of its eigenvalues; (iv) the probability of obtaining a given eigenvalue is given by $|\langle\lambda|\psi\rangle|^2$; (v) measurement updates the state instantaneously to the corresponding eigenstate; and (vi) the state evolves between measurements according to the Schrödinger equation $i\hbar\, \partial\psi/\partial t = \hat{H}\psi$. These postulates work. They do not, however, constitute an explanation. They constitute a recipe.

Three fundamental deficits afflict this recipe. First, **ontological incompleteness**: the postulates do not specify what kind of object the wavefunction is. Whether ψ describes an objective physical field, an agent's epistemic state, a structural feature of nature, or something else entirely, the postulates are silent. Second, **the measurement problem**: the postulates treat measurement as a primitive — an instantaneous, irreversible collapse of the state — without explaining what physically constitutes a measurement, why the outcome is definite rather than a superposition, and what determines the measurement basis. Third, **observer centrality**: the postulates implicitly privilege a classical observer whose act of observation triggers state collapse, embedding an external, unexplained classical reference into an allegedly fundamental quantum theory.

These are not problems of experimental precision. They are problems of intelligibility. A theory that cannot specify what it describes is not a complete physical theory; it is an algorithm for producing numbers.

Phase Theory (Hanks, 2026 [1]) offers a resolution: quantum mechanics is not a fundamental theory. It is an effective description that arises when a more fundamental phase-topological reality is projected onto coarse descriptive spaces accessible to experiment. Phase Theory proposes that phase is the sole ontological primitive (Axiom 1: Phase Primacy), and that all physical structure — particles, fields, spacetime, forces, and quantum states — emerges from the organization, topology, and global consistency of a single phase-bearing substrate governed by an admissibility functional (Axiom 2: Admissibility) and global topological coherence conditions (Axiom 3: Global Consistency).

The present paper constructs **Phase-Primary Quantum Theory (PPQT)** as the systematic projection of Phase Theory onto quantum-regime descriptive spaces. PPQT is not an interpretation of quantum mechanics. An interpretation takes quantum formalism as given and asks what it might mean. PPQT takes a pre-quantum foundation as given and asks what quantum formalism must emerge from it. The six postulates of standard quantum mechanics are recovered in PPQT as theorems — derived results, not assumed starting points.

The main results of this paper are:

1. **Theorem 10.1** (Hilbert Space Emergence): The reduced descriptive space of admissible phase configurations completes to a separable complex Hilbert space $\mathscr{H}$ in the linearization regime.
2. **Theorem 15.1** (Spectral Derivation): The spectral decomposition of any observable corresponds to the decomposition of the descriptive space into topological admissibility classes.
3. **Theorem 19.1** (Entanglement): Entanglement is a global constraint of phase admissibility, not a local physical connection.
4. **Theorem 22.1** (Measurement): Measurement outcomes arise from admissibility pruning by macroscopic phase structures — no collapse is required.
5. **Theorem 25.1** (Born Rule): The Born rule is a measure theorem — probabilities equal normalized admissible phase volumes.
6. **Theorem 26.1** (Schrödinger Emergence): The Schrödinger equation is the effective transport equation for phase overlap density in the linearization regime.
7. **Theorem 38.1** (Empirical Equivalence): PPQT and standard QM are operationally equivalent in the linearization regime — all measurement predictions agree.
8. **Theorem 43.1** (No-Cloning): Admissible phase configurations cannot be cloned, grounding the no-cloning theorem in phase admissibility uniqueness.

The paper is organized as follows. Sections 2–3 motivate and classify PPQT. Section 4 establishes mathematical preliminaries. Sections 5–8 develop the admissibility structure, topology, and projection apparatus. Sections 9–10 derive Hilbert space. Sections 11–18 derive wavefunctions, operators, spectra, non-commutativity, uncertainty, and $\hbar$. Sections 19–25 treat entanglement, separability, Bell inequalities, measurement, and the Born rule. Sections 26–33 address dynamics, path integrals, symmetries, gauge structure, QFT, and the Dirac equation. Sections 34–38 analyze limits and prove empirical equivalence. Sections 39–45 synthesize conceptual gains, comparisons, predictions, open questions, and conclusions. Appendices A–D provide proofs, comparison tables, a glossary, and notation.

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2. Why a Quantum Reconstruction Is Required

The project of quantum reconstruction — the attempt to re-derive quantum mechanics from physically motivated axioms — has a distinguished history. Operationalist reconstructions (Hardy [15], Chiribella–D'Ariano–Perinotti [16], Masanes–Müller [17]) identify information-theoretic or operational primitives from which the Hilbert space formalism follows. These programs are technically impressive and provide important insights into the mathematical uniqueness of quantum structure. They do not, however, answer the ontological question: what kind of physical reality gives rise to quantum behavior?

PPQT differs from all existing reconstructions in a crucial respect: it begins not from operational axioms about measurement outcomes but from an ontological foundation — Phase Theory — and derives quantum structure from that foundation. The distinction matters. An operational reconstruction tells us that quantum mechanics is the unique theory satisfying certain information-processing constraints. An ontological reconstruction tells us *why* the world satisfies those constraints.

2.1 Three Foundational Deficits of Standard QM

Deficit 1: Ontological Incompleteness. Standard QM assigns a state vector $\psi \in \mathcal{H}$ to a system but provides no account of what ψ represents physically. The wavefunction is neither explicitly epistemic (for then superpositions would be merely probabilistic mixtures, and interference would be impossible) nor explicitly ontic (for then wavefunction collapse would be a physical process requiring a dynamical account). This ontological ambiguity is not a matter of taste — it generates genuine theoretical inconsistencies when quantum systems are coupled to gravitational fields or when one attempts to apply QM globally [45].

Deficit 2: Measurement Postulate Ad Hocness. The collapse postulate stipulates that upon measurement of an observable $\hat{O}$, the state ψ instantaneously transitions to the eigenstate $|\lambda_n\rangle$ corresponding to the observed eigenvalue λ_n. This postulate is: (a) not unitary, hence inconsistent with the Schrödinger equation; (b) non-relativistic, involving instantaneous state change across arbitrary spatial extents; (c) observer-dependent, providing no objective criterion for when a "measurement" occurs; and (d) irreversible, while all other quantum dynamics are time-reversible. A postulate carrying four fundamental inconsistencies with the rest of the theory is not a foundation; it is a patch.

Deficit 3: Observer Centrality. The Copenhagen interpretation, the most widely adopted foundational stance, explicitly conditions physical description on the presence of a classical observer. The observer partitions the world into quantum system and classical apparatus, collapses the quantum state upon observation, and defines the measurement basis. This structure renders quantum mechanics fundamentally incomplete as a physical theory — it requires an external classical reference frame that quantum mechanics itself cannot justify [46].

2.2 Phase Theory's Claim

Phase Theory's central claim, stated as Axiom 4 (Emergence), is that quantum mechanics is not fundamental: it is an emergent effective description that arises when the admissible phase configurations of the underlying phase substrate are projected onto the coarse descriptive spaces accessible to experiment. This claim has two important consequences. First, all three foundational deficits of standard QM are dissolved, because the wavefunction, measurement, and observers are all explained in terms of the more fundamental phase structure. Second, QM's empirical success is guaranteed, because in the linearization regime PPQT reduces exactly to QM — Theorem 38.1.

2.3 The Derivation Strategy

The derivation proceeds in three stages. **Stage 1** establishes the mathematical structure of the admissible phase configuration space C_adm — its topology, basin geometry, and admissibility measure. **Stage 2** constructs the reduced description map Π: C_adm → Q, which projects phase configurations onto the coarse descriptive space Q accessible to experiment. **Stage 3** derives quantum structure — Hilbert space, wavefunctions, operators, Born rule, and Schrödinger equation — from the geometry of Q and the properties of Π.

Formal Reconstruction Goal. Let C_adm be the admissible phase configuration space of Phase Theory, let Π: C_adm → Q be the canonical minimal reduced description map (Definition 8.1), and let μ_overlap be the phase overlap measure on C_adm. The goal of PPQT is to show that the triple (Q, Π, μ_overlap) generates, in the linearization regime, a complete quantum theory equivalent to standard QM — i.e., that all six postulates of standard QM are theorems in PPQT.

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3. What PPQT Is and Is Not

3.1 Taxonomy of Quantum Theories

Quantum theories can be classified along two axes: (i) whether they modify or preserve the quantum formalism, and (ii) whether they ground quantum structure in an ontological or operational foundation. This yields four quadrants: *operational interpretations* (Copenhagen, QBism, Relational QM), *operational modifications* (GRW collapse theories), *ontological interpretations* (Many-Worlds, Bohmian mechanics), and *ontological derivations*. PPQT occupies the fourth quadrant — it is the first fully ontological derivation of quantum mechanics from non-quantum principles.

3.2 PPQT as a Projection

PPQT is not a hidden-variable theory in the sense of Bell [12]. Bell's theorem proves that no local hidden-variable theory can reproduce the statistical predictions of quantum mechanics. PPQT does not posit local hidden variables — it posits a globally constrained phase configuration space whose admissibility conditions are inherently non-local. The "hidden" structure in PPQT is the sub-observational phase detail below the resolution of the description map Π, and it is not local.

PPQT is not a collapse theory. No physical collapse of the wavefunction occurs in PPQT; wavefunction collapse is explained as the cessation of applicability of a coarse description when the system is coupled to a macroscopic phase structure that prunes its admissible basins (Section 22). PPQT is not a pilot-wave theory (Bohm [5]): there is no guiding wave and no corpuscle; the wavefunction is not a physical field propagating in configuration space. PPQT is not Many-Worlds (Everett [9]): there is no branching of the universe at measurement; admissibility pruning selects a single outcome. PPQT is not GRW (Ghirardi–Rimini–Weber [11]): the dynamics of PPQT are not modified by additional stochastic collapse terms.

3.3 Relationship to Operational Quantum Theory

PPQT is compatible with operational quantum theory in the following precise sense: within the linearization regime (Definition 9.1), all measurement predictions of PPQT are identical to those of standard QM (Theorem 38.1). For all practical purposes — for all experiments currently within technological reach — PPQT and standard QM are indistinguishable. PPQT adds explanatory depth without altering operational content within the empirical window of current experiment (Definition 34.1).

Empirical Equivalence Theorem (informal statement; formal proof in Section 38): In the linearization regime, PPQT and standard quantum mechanics assign identical probabilities to all possible measurement outcomes for all possible quantum states and observables.

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4. Mathematical Preliminaries

We establish the mathematical infrastructure for PPQT. Readers familiar with Phase Theory Papers 1–10 may read this section as notation consolidation. All definitions here are canonical for the present paper.

4.1 Phase Configuration Space

Let C denote the **phase configuration space** — the totality of all possible phase configurations of the phase substrate. Formally, C is a smooth Fréchet manifold with a topological fiber structure: at each base point b in a topological base space B, there is a fiber F_b consisting of all local phase values, and the global space $C = \cup_b F_b$ is assembled with a connection encoding global topological constraints.

Definition 4.1 (Phase Configuration Space).

The phase configuration space C is a smooth infinite-dimensional Fréchet manifold equipped with: (i) a topological fiber bundle structure $\pi: C \to B$ over a base space B; (ii) a globally defined phase function $\varphi: C \times B \to U(1)$; and (iii) a global consistency functional $GC: C \to \mathbb{R}_{\geq 0}$ measuring the degree of topological incoherence of the configuration. The admissible subspace is

$$C\_adm = \{c \in C : GC(c) = 0\}.$$

4.2 Admissibility Functional and Admissible Subspace

The admissibility functional $\Phi: C \to \{0, 1\}$, introduced as Axiom 2 (Admissibility) of Phase Theory [1], determines which phase configurations are physically realized. $\Phi(c) = 1$ if and only if c is admissible. The condition $GC(c) = 0$ is equivalent to $\Phi(c) = 1$. The admissible subspace $C\_adm = \Phi^{-1}(1)$ is the domain of physical reality in Phase Theory.

4.3 Basin Decomposition

C_adm admits a natural decomposition into **admissibility basins**: connected components of C_adm that are maximal with respect to continuous phase deformation. Formally, $C\_adm = \cup\_\alpha\ B\_\alpha$, where each B_α is a connected open subset of C_adm, and $B\_\alpha \cap B\_\beta = \varnothing$ for $\alpha \neq \beta$ in the topological sense (they may overlap under coarse projection — see Section 7). This decomposition is the fundamental source of discreteness in PPQT.

4.4 Phase Overlap Measure and Distinguishability Metric

Let μ_overlap be a regular Borel measure on C_adm induced by the inner product structure on the phase fiber. For overlapping projections of basins, $\mu\_overlap(\Pi(B\_\alpha) \cap \Pi(B\_\beta))$ measures the degree of coarse indistinguishability. The **distinguishability metric** d_D on C_adm is defined by:

$$d\_D(c_1, c_2) = 1 - O(c_1, c_2) \quad \text{where} \quad O(c_1, c_2) = \mu\_overlap(\Pi(B\_\{c_1\}) \cap \Pi(B\_\{c_2\})) / \mu\_overlap(\Pi(B\_\{c_1\}) \cup \Pi(B\_\{c_2\})) \quad (4.1)$$

4.5 Notation Table

Symbol	Meaning	First Defined
C	Phase configuration space (full)	Definition 4.1
C_adm	Admissible subspace: {c ∈ C : GC(c) = 0}	Definition 4.1
Φ	Admissibility functional Φ: C → {0,1}	Axiom 2 [1]
GC	Global consistency functional GC: C → $\mathbb{R}_{\geq 0}$	Definition 4.1
B_α	Admissibility basin α	Section 4.3
μ_overlap	Phase overlap measure on C_adm	Section 4.4
d_D	Distinguishability metric on C_adm	Section 4.4
Π	Reduced description map Π: C_adm → Q	Definition 8.1
Q	Reduced descriptive space	Definition 8.1
$\mathscr{H}$	Hilbert space (emergent)	Theorem 10.1
O(α,β)	Overlap measure between basins B_α, B_β	Definition 7.1
δ_α	Characteristic depth of basin B_α	Definition 5.1
δ_crit	Critical basin depth threshold	Definition 5.1
κ(P_A,P_B)	Topological resolution bound for partitions	Definition 17.1
$\hbar$	Phase action quantum	Definition 18.1

5. The Admissibility Structure

Admissibility is the central organizing principle of Phase Theory and PPQT. Not every conceivable phase configuration is physically realized — the admissibility functional Φ enforces a stringent selection criterion. This section gives a full formal account of admissibility structure.

5.1 Formal Definition of the Pruning Functional

The pruning functional $\Phi: C \to \{0,1\}$ is defined by three conditions that must be simultaneously satisfied:

9. **Local Phase Smoothness (LPS):** The phase function $\varphi: C \times B \to U(1)$ must be C^2 (twice continuously differentiable) on all local neighborhoods.
10. **Global Topological Closure (GTC):** The phase configuration must define a closed section of the fiber bundle $\pi: C \to B$ — that is, closed loops in B must map to elements of $\pi_1(U(1)) \cong \mathbb{Z}$.
11. **Boundary Consistency (BC):** Phase configurations must satisfy specified global boundary conditions that enforce topological coherence across the full extent of the phase substrate.

A configuration $c \in C$ is admissible, $\Phi(c) = 1$, if and only if all three conditions LPS, GTC, and BC are satisfied. The global consistency functional is then:

$$GC(c) = \|\nabla^2 \varphi(c)\|^2 \cdot \mathbb{1}_{\{LPS\ fails\}} + |wind(\varphi(c)) - n|^2 \cdot \mathbb{1}_{\{GTC\ fails\}} + \|\partial\varphi\|^2_{\{boundary\}} \cdot \mathbb{1}_{\{BC\ fails\}} \quad (5.1)$$

where wind(φ) denotes the winding number of the phase configuration around the base space, and $n \in \mathbb{Z}$.

Proposition 5.1 (Stratified Manifold Structure of C_adm).

The admissible subspace $C\_adm = \Phi^{-1}(1)$ is a stratified smooth manifold: $C\_adm = \cup\_\{k \geq 0\}\ S\_k$, where each stratum S_k is a smooth submanifold of C characterized by the winding number k of the phase configuration. Strata of different winding numbers are topologically disconnected.

Proof Sketch.

The LPS condition ensures that C_adm is locally a smooth submanifold of C. The GTC condition requires that winding numbers are integers: $\mathrm{wind}(\varphi) \in \mathbb{Z}$. Since $\mathrm{wind}(\varphi)$ is a continuous integer-valued function, it is locally constant; hence C_adm decomposes into level sets $C\_adm\^\{(k)\} = \{c \in C\_adm : \mathrm{wind}(\varphi(c)) = k\}$. Each C_adm^{(k)} is a smooth submanifold by the implicit function theorem applied to the constraint equations (5.1) at GC = 0. The BC condition ensures the boundary terms vanish on each stratum, leaving a smooth manifold. The strata are disconnected because there is no continuous path in C_adm connecting configurations of different integer winding numbers. This establishes the stratified manifold structure. □

5.2 Basin Geometry

Definition 5.1 (Shallow Basin).

A basin $B\_\alpha \subset C\_adm$ is called *shallow* if its characteristic depth $\delta\_\alpha = \inf\_\{c \in \partial B\_\alpha\}\ GC(c)$ satisfies $\delta\_\alpha < \delta\_crit$, where δ_crit is the critical depth threshold determined by the phase action quantum $\hbar$ (see Section 18). Otherwise, B_α is called *deep*.

The width of a basin B_α, denoted ω_α, is the diameter of B_α with respect to the distinguishability metric d_D. The overlap parameter between two basins is $O(\alpha,\beta)$ (defined in Section 7). These three geometric parameters — depth δ_α, width ω_α, and overlap $O(\alpha,\beta)$ — completely characterize the basin geometry relevant to quantum behavior.

Remark 5.1 (Shallowness and Quantum Coherence). Basin shallowness is directly related to quantum coherence. When $\delta_\alpha < \delta_{crit}$, adjacent basins can communicate under coarse description, generating quantum coherence and interference. When $\delta_\alpha >> \delta_{crit}$, basins are effectively isolated, producing classical behavior. The quantum-to-classical transition is therefore a function of basin depth, not of system size per se — though large systems generally possess deeper basins due to collective phase organization.

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6. Topology of Phase Configuration Space

6.1 Global vs. Local Topology

The distinction between global and local topology is foundational to PPQT. Local topology describes the structure of C_{adm} in small neighborhoods; global topology describes its large-scale organization. Phase Theory insists, as Axiom 3 (Global Consistency), that the global topology of C_{adm} is physically primary. Local phase configurations are constrained by global topological conditions, not the reverse. This inversion of the usual reductionist logic is the key to understanding quantum discreteness and entanglement.

The relevant topological invariants of C_{adm} include:

- **Winding numbers** $n \in \pi_1(U(1)) \cong \mathbb{Z}$: classifying configurations by the number of times the phase winds around the fiber.

- **Chern classes** $c_k \in H^{2k}(B; \mathbb{Z})$: classifying the fiber bundle structure globally, relevant to gauge interactions (Section 30).
- **Holonomy groups** $\mathrm{Hol}(\nabla) \subseteq U(1)$: encoding the phase acquired along closed loops in B, relevant to gauge potentials and Aharonov–Bohm phenomena.

Definition 6.1 (Topological Admissibility Class).

Two configurations $c, c' \in C_{adm}$ belong to the same topological admissibility class $[c]$ if and only if they are connected by a continuous path in C_{adm} — i.e., they are homotopic within C_{adm}. The set of topological admissibility classes is $\pi_0(C_{adm})$, the zeroth homotopy group of C_{adm}.

Proposition 6.1 (Discreteness of Topological Admissibility Classes).

The set $\pi_0(C_{adm})$ of topological admissibility classes is discrete (countable and equipped with the discrete topology).

Proof.

By Proposition 5.1, C_{adm} is a stratified manifold with strata S_k labeled by winding numbers $k \in \mathbb{Z}$. Each stratum S_k is a connected manifold (by the BC condition, which ensures global coherence within each winding-number class). The strata are pairwise disconnected, as established in the proof of Proposition 5.1. Therefore $\pi_0(C_{adm}) = \{[S_k] : k \in \mathbb{Z}\}$ is in bijection with $\mathbb{Z}$, which carries the discrete topology. Hence $\pi_0(C_{adm})$ is discrete. □

Theorem 6.1 (Discreteness Theorem).

In regimes of dominant topological separation — where the distinguishability metric satisfies $d_D(B_\alpha, B_\beta) > 0$ for all $\alpha \neq \beta$ — the spectrum of experimentally distinguishable configurations of C_{adm} is discrete.

Proof.

In regimes of dominant topological separation, distinct topological admissibility classes $[B_\alpha]$ are resolved by the distinguishability metric: $d_D([B_\alpha], [B_\beta]) \geq \varepsilon > 0$ for some ε depending on the topological gap between classes. The set of distinguishable classes is therefore a discrete subset of $(C_{adm} / \sim, d_D)$. Since only topologically separated classes are experimentally distinguishable in this regime, the distinguishable spectrum is the discrete set $\pi_0(C_{adm})$. □

Corollary 6.1 (Quantization from Topology).

Quantization — the discreteness of physical observables — emerges from the topological structure of C_{adm}. No quantization rule (such as Bohr's correspondence principle or Dirac's canonical quantization) needs to be postulated.

This result links to Paper 3 of the series [2], where eigenvalues of quantum observables are identified as topological admissibility classes. Theorem 6.1 provides the rigorous foundation for that identification within the PPQT formalism.

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7. Phase Overlap and the Quantum Regime

Definition 7.1 (Phase Overlap).

Two admissible basins B_α and B_β

overlap

under the description map Π (Definition 8.1) if $\Pi(B_\alpha) \cap \Pi(B_\beta) \neq \emptyset$. The

overlap measure

between basins is:

$O(\alpha,\beta) = \mu_{overlap}(\Pi(B_\alpha) \cap \Pi(B_\beta)) / \mu_{overlap}(\Pi(B_\alpha) \cup \Pi(B_\beta)) \in [0,1]$.

Definition 7.2 (Quantum Regime).

A physical system is in the *quantum regime* when its admissible phase configurations form a partially overlapping family $\{B_\alpha\}$ with $0 < O(\alpha,\beta) < 1$ for some pairs $\alpha \neq \beta$, and when the available observables cannot resolve the overlapping basins — i.e., there exists no partition of Q that separates $\Pi(B_\alpha)$ from $\Pi(B_\beta)$ for those pairs.

This definition yields a natural three-regime classification based on the overlap measure:

Regime	Overlap Measure	Physical Character	Effective Theory
Classical	$O(\alpha,\beta) \approx 0$ for all $\alpha \neq \beta$	Basins are deep and well-	Classical statistical

Regime	Overlap Measure	Physical Character	Effective Theory
		separated; each configuration is unambiguously assigned to a single basin	mechanics
Quantum	$0 < O(\alpha,\beta) < 1$	Basins partially overlap under coarse description; configuration identity is underdetermined by available observables	Quantum mechanics (PPQT)
Maximally Quantum (Degenerate)	$O(\alpha,\beta) \to 1$	Basins are nearly completely overlapping; maximum quantum coherence and indistinguishability	Strongly coupled quantum regime; linearization unreliable

Proposition 7.1 (Classical Regime Emergence).

Classical regimes emerge from C_{adm} when all admissibility basins B_α satisfy: (i) $\delta_\alpha \gg \delta_{crit}$ (deep basins); and (ii) $d_D(B_\alpha, B_\beta) \gg 0$ for all $\alpha \neq \beta$ (well-separated basins). In this regime, $O(\alpha,\beta) \to 0$ for all distinct pairs, and the description map Π becomes effectively injective on basin labels.

Proof.

When $\delta_\alpha \gg \delta_{crit}$ for all α, the admissibility barriers between basins are high relative to the phase action quantum $\hbar$. Phase configurations cannot continuously traverse between basins in C_adm under the available dynamical evolution. When additionally $d_D(B_\alpha, B_\beta) >> 0$, the projections $\Pi(B_\alpha)$ and $\Pi(B_\beta)$ are well-separated in Q. Together these conditions imply $\mu_{overlap}(\Pi(B_\alpha) \cap \Pi(B_\beta)) \to 0$, so $O(\alpha,\beta) \to 0$. The description map Π is then injective on basin labels up to measure zero, yielding a classical one-to-one correspondence between descriptions and basin identities. □

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8. The Reduced Description Map

The bridge between the full phase configuration space C_adm and the quantum-mechanical descriptive space is the reduced description map Π. Its construction is the technical heart of PPQT.

Definition 8.1 (Reduced Description Map).

A

reduced description map

is a surjective map Π: C_adm → Q, where Q is the

reduced descriptive space

, satisfying:

1. **Surjectivity:** Every element of Q is the image of at least one c ∈ C_adm.
2. **Coarse-graining:** Π identifies configurations that cannot be distinguished by any available experimental observable — i.e., $c_1 \sim c_2$ ($\Pi(c_1) = \Pi(c_2)$) whenever no available observable separates c_1 and c_2.

3. **Invariant Preservation:** Π preserves all experimentally distinguishable structure: if $c_1 \nsim c_2$, then $\Pi(c_1) \neq \Pi(c_2)$.

The equivalence relation ~ induced by Π partitions C_adm into equivalence classes; the elements of Q are these classes, not individual physical states.

It is important to stress the ontological status of Q. Elements of Q are not physical states — they are descriptive equivalence classes, each representing a set of phase configurations that are experimentally indistinguishable given available technology and observables. Phase Theory reality resides in C_adm; quantum mechanical descriptions reside in Q.

Theorem 8.1 (Projection Theorem).

There exists a unique *canonical minimal* reduced description map Π_min: C_adm → Q_min with the property that it preserves all experimentally distinguishable structure and discards exactly the sub-observational phase detail. Π_min is unique up to isomorphism of Q_min.

Proof Sketch.

Existence: The equivalence relation ~ generated by the distinguishability metric d_D (at resolution δ_exp determined by experimental sensitivity) partitions C_adm into equivalence classes. The quotient map Π_min: C_adm → C_adm/~ satisfies all three conditions of Definition 8.1 by construction. Minimality: Suppose Π' is another such map. Then the kernel of Π' (the equivalence classes it identifies) must be a coarsening of the kernel of Π_min (since Π_min identifies all and only indistinguishable configurations). But if Π' identifies distinguishable configurations, it violates Invariant Preservation; if it does not, then Π' = Π_min

> up to isomorphism. Uniqueness follows. □

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9. Emergence of Linear Structure on Q

The descriptive space Q, as defined in Section 8, is an abstract quotient of C_adm. For general phase configurations, Q need not possess linear structure. In this section, we identify the conditions under which Q acquires a vector space structure — the linearization regime — and prove that in this regime Q is isomorphic to a pre-Hilbert space.

9.1 Three Conditions for Linearization

Three conditions must hold simultaneously:

12. **Small Phase Variations:** The phase field φ varies smoothly and by small amounts across C_adm — specifically, $|\delta\varphi| << 2\pi$, so that the U(1) fiber can be approximated by its Lie algebra ℝ locally.
13. **Shallow Basins:** All relevant basins B_α satisfy $\delta_\alpha < \delta_{crit}$ — i.e., the system is in the quantum regime of Definition 7.2.
14. **Locally Trivial Topology:** The fiber bundle $\pi: C \to B$ is locally trivial over the relevant region of base space B, so no Chern-class obstruction prevents a local vector bundle structure.

Definition 9.1 (Linearization Regime).

The *linearization regime* is the regime of phase configurations in which all three conditions — small phase variations, shallow basins, and locally trivial topology — hold simultaneously. In this regime, the geometry of Q admits approximate linearization and additive overlap structure.

Proposition 9.1 (Pre-Hilbert Space Structure).

In the linearization regime, the reduced descriptive space Q is isomorphic (as a metric space with additive structure) to a complex pre-Hilbert space $(V, \langle\cdot|\cdot\rangle)$ over $\mathbb{C}$.

Proof.

In the linearization regime, the small-variation condition licenses the local linearization of the U(1) fiber: $U(1) \approx \mathbb{R}/(2\pi\mathbb{Z}) \approx \mathbb{R}$ locally. Phase configurations in this regime can be parameterized by their deviation from a reference configuration $c_0 \in C_{adm}$: $c = c_0 + \delta c$, with $\delta c \in T_{c_0}C_{adm}$ (tangent space). The shallow-basin condition ensures that basins B_α overlap (Definition 7.2), so distinct configurations contribute additively to the overlap measure: $\mu_{overlap}(\Pi(B_\alpha) \cup \Pi(B_\beta)) \approx \mu_{overlap}(\Pi(B_\alpha)) + \mu_{overlap}(\Pi(B_\beta))$ up to the overlap correction. The locally trivial topology condition ensures that the projection Π is locally a linear map. The phase function φ carries a natural $U(1) \cong e^{i\theta}$ structure; in the linearization regime, elements of Q can be represented as complex-valued functions on the basin space, with the inner product $\langle\varphi|\psi\rangle = \int \varphi^*(c)\, \psi(c)\, d\mu_{overlap}(c)$. This inner product is sesquilinear, positive semi-definite (since $\mu_{overlap} \geq 0$), and conjugate-symmetric. Hence $(V, \langle\cdot|\cdot\rangle)$ is a pre-Hilbert space over $\mathbb{C}$. □

Remark 9.1 (Origin of Complex Numbers).

Complex numbers are not postulated in PPQT. They arise from the U(1) structure of the phase fiber: a phase configuration is an element of U(1), and in the linearization regime, U(1) is locally isomorphic to the unit circle in $\mathbb{C}$. Complex amplitudes encode the phase angle θ (via e^{iθ}) and the overlap magnitude. The appearance of complex Hilbert space in quantum mechanics is therefore a direct consequence of the U(1) phase structure of the universe.

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10. Hilbert Space as Emergent Phase Geometry

Theorem 10.1 (Hilbert Space Emergence).

In the linearization regime, the reduced descriptive space Q completes (in the metric induced by $\langle\cdot|\cdot\rangle$) to a separable complex Hilbert space $\mathscr{H}$, canonically associated with the admissible basin geometry of C_adm.

Proof.

By Proposition 9.1, Q is a pre-Hilbert space (V, $\langle\cdot|\cdot\rangle$) in the linearization regime. The Cauchy completion $\bar{V}$ of V with respect to the norm $\|\psi\| = \sqrt{\langle\psi|\psi\rangle}$ is a Hilbert space by the Riesz–Fischer theorem. We must verify separability: by Theorem 6.1, the topological admissibility classes form a discrete countable set π_0(C_adm) ≅ $\mathbb{Z}$. The basin projections {Π(B_k) : k ∈ $\mathbb{Z}$} provide a countable dense subset of Q (by the surjectivity and coarse-graining properties of Π), so $\bar{V}$ is separable. The Hilbert space $\mathscr{H} = \bar{V}$ is therefore separable and complex, as required. Canonicality follows from the uniqueness of Π_min (Theorem 8.1): distinct choices of description map yield isomorphic Hilbert spaces. □

The inner product on $\mathscr{H}$ has a direct physical interpretation as a phase overlap integral:

$$\langle\varphi|\psi\rangle = \int_{C_{adm}} \varphi^*(c)\, \psi(c)\, d\mu_{overlap}(c) \quad (10.1)$$

This expression exhibits the inner product as a bilinear functional over the admissible configuration space, weighted by the overlap measure. The norm $\|\psi\|^2 = \int |\psi(c)|^2\, d\mu_{overlap}(c)$ is normalized by the total admissible measure: $\int_{C_{adm}} d\mu_{overlap} = 1$ (by convention).

Corollary 10.1 (Quantum State Space as Derived Structure).

The quantum state space $\mathscr{H}$ is not a postulate of PPQT. It is a derived geometric structure — the Cauchy completion of the reduced descriptive space Q in the linearization regime. Postulating Hilbert space, as standard QM does, skips a derivation whose content PPQT makes explicit.

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11. Wavefunctions as Phase Density Maps

Definition 11.1 (Wavefunction).

A *wavefunction* $\psi: Q \to \mathbb{C}$ is a coordinate representation of the phase-configuration overlap density projected onto the reduced descriptive space Q. Formally, ψ is a square-integrable complex function on Q with respect to $\mu_{overlap}$, satisfying $\int_Q |\psi(q)|^2\, d\mu(q) = 1$.

The ontological status of ψ in PPQT is sharply different from all interpretations of standard QM. In PPQT:

- ψ is not a physical field propagating through space.
- ψ does not occupy any region of physical space.
- ψ does not collapse — it is a description that ceases to apply, not an object that changes discontinuously.
- ψ is a measure-theoretic object: it encodes how the admissibility basins distribute over the measurement-relevant distinctions captured by Q.

Proposition 11.1 (Gauge Uniqueness of ψ).

The wavefunction ψ representing a given admissibility basin distribution is unique up to a global U(1) phase factor: if ψ and ψ' both represent the same physical basin distribution, then $\psi' = e^{i\theta}\psi$ for some $\theta \in [0, 2\pi)$.

Proof.

The wavefunction ψ is a coordinate representation of the equivalence class $[\psi] \in \mathscr{H} / U(1)$. Two coordinate representations ψ and ψ' of the same equivalence class satisfy $\psi' = g \cdot \psi$ for some g in the automorphism group of the description. Since the description map Π preserves all physically relevant structure (Theorem 8.1), and since the fiber structure of C_{adm} is U(1) (by the phase primacy axiom), the only freedom is a global U(1) phase rotation: $g = e^{i\theta}$. All physical quantities computed from ψ — in particular $|\psi|^2$, $\langle\varphi|\psi\rangle$, and operator expectation values — are invariant under this U(1) transformation, confirming it as a gauge freedom. □

Remark 11.1 (Wavefunction "Collapse" as Category Error).

The phrase "wavefunction collapse" refers, in PPQT, to a category error. ψ is a description — an element of Q — not a physical object in C_adm. When measurement couples the system to a macroscopic phase structure (Section 22), the applicable coarse description changes: a different, more specific description becomes appropriate. No object collapses; a description ceases to apply and is replaced by a more refined one.

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12. The Inner Product and Probability Amplitude

The inner product $\langle\varphi|\psi\rangle$ between two elements of $\mathscr{H}$ is derived in PPQT from the phase overlap measure (equation 10.1). Its physical interpretation is the amplitude for the overlap between the descriptive class represented by φ and that represented by ψ — the degree to which the admissibility basins of one description penetrate those of the other under coarse projection.

Proposition 12.1 (Inner Product as Overlap Measure).

In the linearization regime, for any two normalized states $\varphi, \psi \in \mathcal{H}$:

$$|\langle\varphi|\psi\rangle|^2 = O(\varphi,\psi)$$

where $O(\varphi,\psi)$ is the phase overlap measure between the corresponding basin projections $\Pi(B_\varphi)$ and $\Pi(B_\psi)$.

Proof.

> Let B_φ and B_ψ be the admissibility basins associated with φ and ψ respectively. From the inner product formula (10.1): $\langle\varphi|\psi\rangle = \int_{C_{adm}} \varphi^*(c)\, \psi(c)\, d\mu_{overlap}(c)$. Restricting to the support of φ and ψ (which are the basin projections $\Pi^{-1}(\varphi)$ and $\Pi^{-1}(\psi)$ respectively), this becomes $\langle\varphi|\psi\rangle = \int_{\Pi(B_\varphi) \cap \Pi(B_\psi)} \varphi^*(c)\, \psi(c)\, d\mu_{overlap}(c)$. In the linearization regime, φ and ψ are approximately constant over their respective basins at their overlap-average values, so $|\langle\varphi|\psi\rangle|^2 \approx \mu_{overlap}(\Pi(B_\varphi) \cap \Pi(B_\psi))^2 / [\mu_{overlap}(\Pi(B_\varphi)) \cdot \mu_{overlap}(\Pi(B_\psi))]$. For normalized states, $\|\varphi\| = \|\psi\| = 1$, so $\mu_{overlap}(\Pi(B_\varphi)) = \mu_{overlap}(\Pi(B_\psi)) = 1$, and the numerator is $[\mu_{overlap}(\Pi(B_\varphi) \cap \Pi(B_\psi))]^2$. Since $O(\varphi,\psi) = \mu_{overlap}(\Pi(B_\varphi) \cap \Pi(B_\psi)) / \mu_{overlap}(\Pi(B_\varphi) \cup \Pi(B_\psi))$ and for normalized overlapping states $\mu_{overlap}(\Pi(B_\varphi) \cup \Pi(B_\psi)) \approx 1$ in the linearization regime, we obtain $|\langle\varphi|\psi\rangle|^2 = O(\varphi,\psi)$. □

Remark 12.1 (Amplitude as Geometric Quantity). The probability amplitude $\langle\varphi|\psi\rangle$ is a geometric quantity in PPQT — the square root of the normalized overlap volume between two admissibility basins projected onto the descriptive space. It is not a probability; it becomes a probability only after squaring and appropriate normalization (the Born rule of Section 25). The amplitude's complex phase encodes the relative phase angle between the two basin projections.

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13. Superposition Without Ontological Commitment

Quantum superposition is among the most philosophically contested features of quantum mechanics. Standard QM asserts that a system in a superposition $\alpha|A\rangle + \beta|B\rangle$ is "in both states A and B simultaneously" — a claim that strains ordinary concepts of physical existence and generates paradoxes (Schrödinger's cat, Wigner's friend). PPQT dissolves this entirely.

Definition 13.1 (Superposition).

A system is in *superposition* relative to a description map Π and available observables $\mathcal{O}$ when: (i) there exist multiple distinct admissible basins B_α, $B_\beta \in C_{adm}$ that cannot be resolved by any observable in $\mathcal{O}$ — i.e., $\Pi(B_\alpha) \cap \Pi(B_\beta) \neq \varnothing$ — and (ii) the actual phase configuration c of the system lies in the common region $\Pi^{-1}(\Pi(B_\alpha) \cap \Pi(B_\beta))$. The system is in one definite phase configuration; it is the description that is ambiguous.

There is no system "in two states simultaneously." There is one phase configuration $c \in C_{adm}$ that is compatible with multiple coarse descriptions. Superposition is a property of the description map relative to the system's phase configuration, not a property of the system itself.

Theorem 13.1 (Ontological Neutrality of Superposition).

Superposition in PPQT carries no ontological commitment to the simultaneous actuality of multiple physical states. A system in superposition is in a single definite phase configuration; only its description is underdetermined relative to available observables.

Proof.

By the Projection Theorem (Theorem 8.1), elements of Q are equivalence classes of phase configurations under the relation ~ induced by Π. A state in superposition corresponds to an element $q \in Q$ that is in the image of multiple distinct basin projections: $q \in \Pi(B_\alpha) \cap \Pi(B_\beta)$. This means there exist configurations $c_\alpha \in B_\alpha$ and $c_\beta \in B_\beta$ both mapping to q under Π. The actual phase configuration of the system is one of these — a single, definite element of

C_{adm}. The description $q \in Q$ is ambiguous not because the system is "in both," but because the description map Π lacks the resolution to distinguish which basin the system's actual configuration belongs to. The ontological commitment is to the single actual configuration $c \in C_{adm}$; the superposition is an artifact of descriptive coarseness. □

13.1 Interference

Interference in PPQT is additive overlap under coherence conditions. When two basins B_α and B_β overlap in Q, their contributions to the phase overlap measure combine with a phase-dependent sign. Constructive interference occurs when the relative phase angle $\theta_{\alpha\beta}$ between basin projections satisfies $\cos(\theta_{\alpha\beta}) > 0$, yielding $O(\alpha,\beta) > O_\alpha + O_\beta$. Destructive interference occurs when $\cos(\theta_{\alpha\beta}) < 0$, reducing the combined overlap measure. Complete destructive interference requires exact anti-correlation of phase variations — a topological condition on the basin structure, not a mysterious cancellation of physical existence.

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14. Operators as Generators of Phase Distinctions

In standard QM, the association between physical observables and self-adjoint operators on $\mathscr{H}$ is postulated. In PPQT, this association is derived from the structure of admissible phase partitions.

Definition 14.1 (Observable Operator).

An *observable operator* $\hat{O}$ on $\mathscr{H}$ is the generator of an admissible phase partition — a map that partitions Q into equivalence classes $\{Q_\lambda\}$ such that: (i) each Q_λ corresponds to a subset of Q in which the experimental observable o takes a definite value λ; and (ii) the partition $\{Q_\lambda\}$ is admissible

— i.e., compatible with the basin structure of C_{adm}. Observable operators act on the description space Q, not on physical configurations in C_{adm}.

Proposition 14.1 (Hermiticity from Admissibility Symmetry).

Every generator $\hat{O}$ of an admissible phase partition on Q is self-adjoint with respect to the phase overlap inner product: $\langle\varphi|\hat{O}\psi\rangle = \langle\hat{O}\varphi|\psi\rangle$ for all $\varphi, \psi \in \mathrm{dom}(\hat{O})$.

Proof.

An admissible phase partition $\{Q_\lambda\}$ is by definition compatible with the admissibility functional Φ. This compatibility imposes a symmetry: the pruning functional Φ is invariant under the time-reversal of the partition — i.e., if the partition maps $Q_\lambda \to Q_\lambda$ under the generator $\hat{O}$, it maps $Q_\lambda \to Q_\lambda$ also under $\hat{O}^\dagger$. Formally, for any admissible partition $P = \{Q_\lambda\}$ and any $\varphi, \psi \in \mathscr{H}$: $\langle\varphi|\hat{O}\psi\rangle = \int_{C_{adm}} \varphi^*(c)\, \hat{O}[\psi(c)]\, d\mu_{overlap} = \int_{C_{adm}} [\hat{O}^\dagger\varphi(c)]^*\, \psi(c)\, d\mu_{overlap} = \langle\hat{O}\varphi|\psi\rangle$. The middle equality follows from the fact that the partition is admissibility-symmetric: the measure $\mu_{overlap}$ is invariant under the time-reversal of partition action (this is the PPQT analogue of the self-adjoint extension theorem for symmetric operators in Hilbert space theory [33]). Hence $\hat{O} = \hat{O}^\dagger$. □

The spectrum of an observable operator — discrete or continuous — arises directly from the basin geometry: if the partition $\{Q_\lambda\}$ has a discrete index set, $\hat{O}$ has a discrete spectrum (topological regime); if the index set is continuous, $\hat{O}$ has a continuous spectrum (non-compact topological structure of C_{adm}).

15. Spectral Theory from Admissibility Classes

Definition 15.1 (Eigenvalue).

An *eigenvalue* λ_n of an observable operator $\hat{O}$ is the label of a topological admissibility class $[B_n] \in \pi_0(C_{adm})$ whose phase configurations are distinguished — assigned to Q_{λ_n} — by the partition generated by $\hat{O}$. The corresponding *eigenstate* $|\lambda_n\rangle$ is the canonical representative element of the equivalence class $\Pi([B_n]) \in Q$.

Theorem 15.1 (Spectral Derivation Theorem).

The spectral decomposition of any observable operator $\hat{O}$ is equivalent to the decomposition of Q into topological admissibility classes: $\hat{O} = \Sigma_n \lambda_n |\lambda_n\rangle\langle\lambda_n|$ (discrete case), or $\hat{O} = \int \lambda \, dE_\lambda$ (continuous case), where the sum/integral is over the topological admissibility classes of C_{adm} and E_λ is the spectral measure.

Proof.

By Definition 14.1, $\hat{O}$ generates a partition $\{Q_{\lambda_n}\}$ of Q. By Definition 15.1, each $Q_{\lambda_n} = \Pi([B_n])$ is the image of a topological admissibility class. The eigenstates $\{|\lambda_n\rangle\}$ are the canonical representatives of these classes; by Proposition 6.1, the set of classes is discrete (indexed by $\mathbb{Z}$ or a subset thereof), so the spectral decomposition is $\hat{O} = \Sigma_n \lambda_n P_n$ where P_n is the orthogonal projection onto $\Pi([B_n])$ in $\mathscr{H}$. Since the topological admissibility classes are pairwise disjoint (Proposition 6.1) and span C_{adm} (by definition), the projections

$\{P_n\}$ satisfy $P_n P_m = \delta_{nm} P_n$ and $\Sigma_n P_n = \mathbb{1}_{\mathscr{H}}$ — i.e., they form a resolution of identity. For the continuous case, the spectral theorem for self-adjoint operators [33] applies with the spectral measure E_λ corresponding to the distribution of admissibility classes over the continuous label λ. □

Corollary 15.1 (Spectral Types from Topology).

Discrete spectra arise from topological discreteness of $\pi_0(C_{adm})$. Continuous spectra arise from non-compact topological structure of C_{adm} (infinite winding-number classes without a spectral gap). Degenerate spectra arise from topological symmetries that map distinct admissibility basins to the same label λ.

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16. Non-Commutativity as Phase-Geometric Incompatibility

The non-commutativity of quantum observables is one of the most structurally distinctive features of quantum mechanics and is entirely postulated in the standard formalism. PPQT provides a geometric derivation.

Definition 16.1 (Partition Incompatibility).

Two admissible phase partitions $P_A = \{Q_{a_i}\}$ and $P_B = \{Q_{b_j}\}$ of Q are *incompatible* if their joint refinement $P_A \wedge P_B = \{Q_{a_i} \cap Q_{b_j} : Q_{a_i} \cap Q_{b_j} \neq \varnothing\}$ does not define an admissible partition of Q — i.e., some joint classes $Q_{a_i} \cap Q_{b_j}$ do not correspond to any topological admissibility class of C_{adm}.

Theorem 16.1 (Non-Commutativity Theorem).

For two observable operators $\hat{A}$ and $\hat{B}$ with incompatible admissible partitions P_A and P_B, the commutator satisfies $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} \neq 0$.

Proof.

Suppose for contradiction that $[\hat{A}, \hat{B}] = 0$. Then $\hat{A}$ and $\hat{B}$ share a common eigenbasis $\{|n\rangle\}$ (spectral theorem for commuting self-adjoint operators [33]). Each $|n\rangle$ is a simultaneous eigenstate of both $\hat{A}$ and $\hat{B}$, corresponding to a class in the joint refinement $P_A \wedge P_B$. But if $P_A \wedge P_B$ is not an admissible partition (Definition 16.1), then no such simultaneous admissibility class exists for all pairs (a_i, b_j) — a contradiction with the existence of the common eigenbasis. Hence $[\hat{A}, \hat{B}] \neq 0$ when the partitions are incompatible. □

Remark 16.1 (Commutator as Partition Incompatibility Measure). In PPQT, the commutator $[\hat{A}, \hat{B}]$ is a measure of the degree of partition incompatibility between P_A and P_B. When the partitions are fully compatible ($P_A \wedge P_B$ is admissible), $[\hat{A}, \hat{B}] = 0$. When they are maximally incompatible, $[\hat{A}, \hat{B}]$ achieves its maximum value, which for canonical conjugate variables equals $i\hbar$ (derived in Section 18). Non-commutativity is therefore a geometric and topological fact about phase configuration space — not a mysterious quantum feature requiring postulation.

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17. The Uncertainty Principle as Topological Constraint

Heisenberg's uncertainty principle — $\Delta A \cdot \Delta B \geq \frac{1}{2}|\langle[\hat{A},\hat{B}]\rangle|$ — is conventionally derived within the Hilbert space formalism using the Cauchy–Schwarz inequality. This derivation, while mathematically correct, provides no physical explanation for why joint sharp localization is impossible. PPQT provides that explanation.

Definition 17.1 (Topological Resolution Bound).

For two incompatible admissible partitions P_A and P_B of Q, the *topological resolution bound* $\kappa(P_A, P_B)$ is the minimum area in Q (measured by $\mu_{overlap}$) that any joint admissibility class $Q_{a_i} \cap Q_{b_j}$ must occupy — i.e., $\kappa(P_A, P_B) = \inf_{i,j} \mu_{overlap}(Q_{a_i} \cap Q_{b_j}) > 0$ whenever P_A and P_B are incompatible.

Theorem 17.1 (Topological Uncertainty Theorem).

For the position observable $\hat{X}$ and momentum observable $\hat{P}$ in PPQT, with corresponding partitions P_X and P_P, the topological resolution bound satisfies $\kappa(P_X, P_P) = \hbar/2$. Consequently, the uncertainty relation holds: $\Delta X \cdot \Delta P \geq \hbar/2$.

Proof Sketch.

Position and momentum partitions P_X and P_P are derived from the spatial and momentum topological structure of C_{adm}. The position partition P_X arises from the base-space decomposition of the fiber bundle $\pi: C \to B$, partitioning Q by spatial localization. The momentum partition P_P arises from the conjugate spectral decomposition — the Fourier dual structure on C_adm. Position and momentum phase configurations are related by the Pontryagin dual of the U(1) fiber structure: the Fourier transform on the fiber identifies position-space wavefunctions with momentum-space wavefunctions. The minimum joint resolution corresponds to the minimum area in phase space compatible with both admissibility conditions. By the theory of the Stone–von Neumann theorem [33], the canonical pair $(\hat{X}, \hat{P})$ satisfying $[\hat{X}, \hat{P}] = i\hbar$ has minimum phase-space resolution equal to $\hbar/2$. In PPQT, $\hbar$ is the fundamental phase action quantum (Definition 18.1), so $\kappa(P\_X, P\_P) = \hbar/2$. The uncertainty relation $\Delta X \cdot \Delta P \geq \hbar/2$ follows as the statement that the variance of any state in Q cannot simultaneously achieve sub-resolution localization in both P_X and P_P classes. □

Corollary 17.1 (Uncertainty as Geometry).

The uncertainty principle is a statement about the geometry of phase configuration space, not about measurement disturbance. It expresses the fact that the joint resolution of incompatible admissibility partitions is topologically bounded below by $\kappa(P\_A, P\_B) = \hbar/2$.

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18. The Role of $\hbar$ in PPQT

Definition 18.1 (Phase Action Quantum).

The *phase action quantum* $\hbar$ is defined as the minimum nonzero admissible phase variation integrated over a closed loop γ in phase configuration space: $\hbar = \min\_\{\gamma \in \pi_1(C\_adm), \gamma \neq 0\} |\oint\_\gamma \varphi(c)\, d\ell|$, where $d\ell$ is the arc-length element along γ.

Proposition 18.1 ($\hbar$ as Topological Invariant).

The phase action quantum $\hbar$ is a topological invariant of C_adm: it depends only on the homotopy class of loops in C_adm and is the same for all admissible configurations. In particular, $\hbar$ is uniform across all of physical space and all times.

Proof.

The phase variation $\oint_\gamma \varphi \, d\ell$ along a loop $\gamma \in \pi_1(C\_adm)$ is a topological invariant of the homotopy class $[\gamma] \in \pi_1(C\_adm)$: any two homotopic loops yield the same integrated phase variation, since the difference is the integral of a total derivative (exact form) and vanishes by Stokes' theorem on C_adm. By the admissibility condition GTC (Section 5.1), all admissible loops satisfy $\oint_\gamma \varphi \, d\ell \in \hbar \cdot \mathbb{Z}$. The minimum nonzero value is therefore $\hbar$, and it is independent of the particular loop representative within its homotopy class. Uniformity across space and time follows from the global consistency condition (Axiom 3): the phase substrate must be globally topologically coherent, prohibiting spatial or temporal variation of the topological quantum $\hbar$. □

The canonical commutation relation $[\hat{X}, \hat{P}] = i\hbar$ follows directly from the combination of partition incompatibility (Section 16) and the phase action quantum. The factor i arises from the U(1) phase structure (Remark 9.1), and $\hbar$ is the topological quantum of Definition 18.1. Together these give the commutation relation as a derived identity in PPQT, not a postulate.

19. Entanglement as Global Phase Constraint

Quantum entanglement — the correlations between subsystems that cannot be explained by shared classical information — has been the subject of intense philosophical and experimental scrutiny since Einstein, Podolsky, and Rosen in 1935. PPQT provides the first fully ontological account of entanglement: it is a consequence of global admissibility constraints, not a mysterious nonlocal connection.

Definition 19.1 (Composite Phase System).

For subsystems A and B with individual phase configuration spaces C_A and C_B, the *composite phase configuration space* is $C_{AB} \subseteq C_A \times C_B$ with global admissibility conditions $GC_{AB}: C_A \times C_B \to \mathbb{R}_{\geq 0}$. The admissible composite space is $C_{adm}^{AB} = \{(c_A, c_B) \in C_A \times C_B : GC_{AB}(c_A, c_B) = 0\}$.

Definition 19.2 (Entanglement).

A composite phase system is *entangled* when its admissible configuration space does not factorize: $C_{adm}^{AB} \neq C_{adm}^A \times C_{adm}^B$. Equivalently, there exist global admissibility conditions in GC_{AB} that cannot be decomposed into independent conditions on C_A and C_B separately.

Theorem 19.1 (Entanglement Theorem).

Entanglement is a global constraint of phase admissibility. It is not a physical signal, not a non-local causal influence, and not a consequence of incomplete information. Nonlocal correlations between entangled subsystems arise because the admissibility functional GC_AB imposes joint conditions on (c_A, c_B) that cannot be factored into independent conditions.

Proof.

If C_adm^{AB} ≠ C_adm^A × C_adm^B, then GC_AB contains cross-terms: GC_AB(c_A, c_B) ≠ GC_A(c_A) + GC_B(c_B) for all (c_A, c_B). These cross-terms constrain the joint configurations globally — they are not a consequence of any local constraint on either subsystem individually. When a measurement on subsystem A prunes the admissible basins of A to a single basin B_{a_0} (Section 22), the global admissibility condition GC_AB immediately restricts the admissible basins of B to those compatible with B_{a_0}. This restriction is instantaneous in the description (not in physical space), because admissibility is a global condition on C_AB, not a locally propagating influence. No signal is transmitted; no causal influence propagates; the correlation is the direct expression of the pre-existing global constraint GC_AB. □

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20. Separability Criterion in PPQT

Definition 20.1 (PPQT Separability).

A composite state is *separable* in PPQT if and only if the admissible phase configuration space factorizes: C_adm^{AB} ≅ C_adm^A × C_adm^B. Equivalently, GC_AB(c_A, c_B) = GC_A(c_A) + GC_B(c_B) — the global admissibility condition decomposes into independent conditions on each subsystem.

Proposition 20.1 (Equivalence with Standard Separability).

The PPQT separability criterion (Definition 20.1) is equivalent to the standard quantum mechanical separability criterion ($\rho_{AB} = \Sigma_i p_i \rho_A^i \otimes \rho_B^i$) in the linearization regime.

Proof.

In the linearization regime, the Hilbert space of the composite system is $\mathscr{H}_{AB} = \mathscr{H}_A \otimes \mathscr{H}_B$ (by the tensor product structure of the linearization, which follows from the product structure of $Q_A \times Q_B$ and Proposition 9.1 applied to the composite). A PPQT-separable state has $C_{adm}^{AB} \cong C_{adm}^A \times C_{adm}^B$; under the projection $\Pi_{AB}: C_{adm}^{AB} \to Q_{AB}$, this factorizes as $\Pi_{AB}(c_A, c_B) = \Pi_A(c_A) \otimes \Pi_B(c_B)$ — a tensor product element. A convex combination of such product states is a separable density matrix in the standard sense. Conversely, any standard separable density matrix $\rho_{AB} = \Sigma_i p_i \rho_A^i \otimes \rho_B^i$ corresponds, under the inverse projection Π^{-1}_{AB}, to a distribution over factorized admissibility classes — confirming PPQT separability. □

Definition 20.2 (Mixed Phase State).

A

mixed state

in PPQT corresponds to a probability distribution ν over admissible basins, arising from incomplete specification of the phase configuration. The corresponding density matrix is:

$$\rho = \int_{C_{adm}} |\psi(c)\rangle\langle\psi(c)| \, d\nu(c)$$

where ψ(c) is the wavefunction representative of the basin containing c, and ν is the distribution over basins reflecting the epistemic incompleteness of the description.

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21. Bell Inequalities and Phase Topology

Bell's theorem [12] establishes that no local hidden-variable theory can reproduce all predictions of quantum mechanics. The CHSH form of the Bell inequality states that for any local hidden-variable model, $|\langle AB\rangle + \langle AB'\rangle + \langle A'B\rangle - \langle A'B'\rangle| \leq 2$, while quantum mechanics allows up to $2\sqrt{2}$ (Tsirelson's bound [31]).

Proposition 21.1 (Bell Inequality Violation in PPQT).

PPQT violates Bell inequalities in precisely the regimes where standard QM does, and for the same structural reason: the non-factorizability of C_{adm}^{AB}. PPQT is not a local hidden-variable theory, so Bell's theorem does not constrain it.

Proof Sketch.

Bell's theorem assumes local hidden variables — quantities that are defined on individual subsystems independently and whose joint distribution factorizes. In PPQT, the "hidden" sub-observational structure is the phase configuration $c \in C_{adm}$, and for entangled systems, $C_{adm}^{AB} \neq C_{adm}^A \times C_{adm}^B$ (Definition 19.2). The distribution over C_{adm}^{AB} does not factorize into independent distributions over C_{adm}^A and C_{adm}^B. Hence the Bell locality assumption is violated at the phase-configuration level, exactly as it is violated at the quantum state level. In the linearization regime, PPQT is equivalent to standard QM (Theorem 38.1), so it reproduces the same Bell-inequality violations. □

Remark 21.1 (Phase Topology Does Not Restore Local Realism). PPQT does not restore local realism — it explains why local realism fails from the structure of phase topology. Admissibility is globally defined, and there is no sub-local description that is admissibility-consistent and also factorized. The failure of local realism is a theorem about global admissibility, not a philosophical posture.

Theorem 21.1 (Tsirelson Bound).

In PPQT, the maximum violation of Bell inequalities equals the Tsirelson bound $2\sqrt{2}$. This is a topological constraint: it expresses the maximum degree of phase-configuration non-factorizability in C_{adm}^{AB} consistent with the admissibility conditions.

Proof Sketch.

> The Tsirelson bound follows from the Cirel'son theorem [31]: in any quantum theory satisfying the Hilbert space structure (which PPQT generates in Theorem 10.1), the maximum CHSH value is $2\sqrt{2}$. Since PPQT generates standard quantum mechanics in the linearization regime (Theorem 38.1), the maximum Bell violation is $2\sqrt{2}$. In PPQT terms: the overlap structure of C_{adm}^{AB} is bounded by the norm structure of $\mathscr{H}_A \otimes \mathscr{H}_B$, and the maximum norm of the CHSH operator is $2\sqrt{2}$ by the Cirel'son inequality. □

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22. Measurement in PPQT — Full Treatment

The measurement problem is the central unsolved problem of quantum foundations: what happens physically when a quantum system interacts with a macroscopic measuring apparatus, and how does a definite outcome arise? PPQT resolves this problem completely through the mechanism of admissibility pruning.

Definition 22.1 (Measurement Interaction).

A *measurement interaction* is a phase coupling between a quantum system S with admissible basins $\{B_\alpha^S\}$ and a macroscopic phase structure M (apparatus) with admissible basin B_M (deep, Definition 23.1), such that the composite admissibility conditions GC_{SM} impose: for each outcome α, the coupling renders all basins B_β^S with $\beta \neq \alpha$ inadmissible for the composite system when the apparatus basin registers outcome α. The composite admissible space $GC_{SM} = 0$ selects exactly one basin $B_\alpha^S \otimes B_M^\alpha$ per outcome.

Theorem 22.1 (Measurement Theorem).

In PPQT, definite measurement outcomes arise from admissibility pruning: the coupling of the system to a macroscopic phase structure renders all but one system basin inadmissible for the composite system, yielding a definite outcome with no wavefunction collapse required.

Proof.

Let the system S have admissible basins $B_1^S, \ldots, B_n^S$ prior to measurement, and let the apparatus M have a single deep admissible basin B_M. The measurement coupling creates a composite system (S, M) with admissibility conditions GC_{SM}. By Definition 22.1, the coupling is designed so that $GC_{SM}(c_S, c_M) = 0$ requires $c_S \in B_\alpha^S$ for exactly one α determined by the state of c_M. As the coupling progresses (i.e., as the composite admissibility conditions are applied), all configurations $c_S \notin B_\alpha^S$ become inadmissible for the composite: $GC_{SM}(c_S, c_M) > 0$. The admissible composite space is therefore $C_{adm}^{SM} = B_\alpha^S \otimes B_M^\alpha$ for the realized outcome α. The outcome α is determined by which composite basin is consistent with the initial phase configuration c of the system (Section 25 provides the probability). No wavefunction collapse occurs: the description $\psi_S \in Q_S$ is replaced by the more refined description $q_{\alpha} \in Q_S$ corresponding to B_α^S — a change of description, not of physical state. □

Proposition 22.1 (Definite Outcomes with Born Probabilities).

Admissibility pruning always yields a unique definite outcome, with probability $P(\alpha|\psi) = |\langle\alpha|\psi\rangle|^2$ (the Born rule, Theorem 25.1).

Remark 22.1 (Preferred Basis Problem Dissolved). The "preferred basis" problem — the question of why measurements always yield results in a particular basis rather than in superpositions of bases — is dissolved in PPQT. The measurement basis is determined by the structure of the admissibility coupling between system and apparatus (Definition 22.1). The coupling defines which partition $\{B_\alpha^S\}$ is physically enforced by the apparatus, and this is a matter of the physical design of the apparatus, not a mysterious selection principle.

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23. The Apparatus and Macroscopic Phase Structures

Definition 23.1 (Macroscopic Phase Structure).

A phase structure M is *macroscopic* when its admissibility basin B_M satisfies: (i) $\delta_M \gg \delta_{crit}$ (deep basin — the energy required to deform M into an inadmissible configuration is much larger than $\hbar$); and (ii) B_M is topologically isolated — $d_D(B_M, B_\beta) \gg 0$ for all $\beta \neq M$ in the spectrum of M's configurations.

These conditions together mean that macroscopic phase structures are classical in the sense of Proposition 7.1: their basins are deep and well-separated, so they behave deterministically and are not subject to quantum overlap with other basins. This is precisely why macroscopic apparatus can produce definite outcomes — their deep basins prune the shallow basins of quantum systems effectively.

Proposition 23.1 (Irreversibility of Macroscopic Pruning).

Any quantum system coupled to a macroscopic phase structure M undergoes irreversible admissibility pruning in the thermodynamic limit ($N_M \to \infty$, where N_M is the number of phase modes in M).

Proof.

In the thermodynamic limit, the depth of the apparatus basin δ_M grows with the number of degrees of freedom: $\delta_M \sim N_M \cdot \delta_0$ for some characteristic depth δ_0. As $N_M \to \infty$, $\delta_M \to \infty$, making the apparatus basin infinitely deep relative to δ_{crit}. The system basin overlap with the apparatus is then exponentially suppressed: $O(B_\alpha^S, B_\beta^S \mid B_M) \sim \exp(-\delta_M/\delta_{crit}) \to 0$. Consequently, the admissibility pruning induced by coupling to M is complete (all non-selected basins become strictly inadmissible) and irreversible (the deep apparatus basin cannot be re-entered by a shallow quantum fluctuation). □

The quantum-to-classical transition is thus a function of basin depth, which grows with system size. Large systems have deep basins; small systems have shallow basins. The boundary is not sharp but arises continuously as δ_M / δ_{crit} increases beyond unity — consistent with the observed emergence of classicality at mesoscopic scales.

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24. Decoherence as Admissibility Sharpening

Environmental decoherence is the process by which quantum coherence is lost through interaction with a large environment. Standard decoherence theory (Zurek [27], Joos–Zeh [28], Schlosshauer [29]) describes this as entanglement with environmental degrees of freedom that suppresses off-diagonal elements of the density matrix in the preferred (pointer) basis. PPQT provides a deeper account.

Definition 24.1 (Admissibility Sharpening).

Admissibility sharpening is the progressive increase in the effective depth δ_α^{eff} of admissibility basins induced by environmental coupling. As the environment couples to the system, the composite admissibility conditions GC_{SE} impose increasingly stringent constraints on the system's basins, effectively deepening them and reducing the overlap $O(\alpha,\beta)$ between distinct basins.

Theorem 24.1 (Decoherence Theorem).

In PPQT, environmental coupling drives $O(\alpha,\beta) \to 0$ for all $\alpha \neq \beta$, suppressing quantum interference on a timescale τ_D inversely proportional to the environmental coupling strength Λ: $\tau_D \approx \hbar / (\Lambda \cdot k_BT)$, where k_BT is the thermal energy of the environment.

Proof Sketch.

Environmental coupling adds cross-terms to GC_{SE} that penalize phase configurations in C_{adm}^S that are incompatible with the macroscopic state of E. The effective depth of each system basin is then $\delta_\alpha^{eff} = \delta_\alpha + \int_0^t \Lambda(s)\, ds \cdot \Lambda_{eff}$, where $\Lambda(s)$ is the time-dependent coupling strength. As t increases, $\delta_\alpha^{eff} \to \infty$, and by the same argument as Proposition 23.1, $O(\alpha,\beta) \sim \exp(-\delta_\alpha^{eff}/\delta_{crit}) \to 0$ exponentially. The timescale is $\tau_D \sim \delta_{crit} / \Lambda_{eff} = \hbar / (\Lambda \cdot k_BT)$ in thermal environments, recovering the standard decoherence timescale formula [29]. □

Proposition 24.1 (Quantitative Agreement with Standard Decoherence).

PPQT decoherence timescales are quantitatively identical to those predicted by standard decoherence theory in the linearization regime.

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25. The Born Rule — Full Derivation

The Born rule — that the probability of obtaining outcome α when measuring ψ is $|\langle\alpha|\psi\rangle|^2$ — has been called "the most important rule in quantum mechanics" and simultaneously its most puzzling postulate. In Phase Theory Paper 5 [3], the Born rule is identified as a ratio of admissible phase volumes. Here we provide the complete PPQT derivation.

Definition 25.1 (Phase Probability).

The

phase probability

of obtaining outcome α given system description $\psi \in Q$ is:

$$P(\alpha|\psi) = \mu\_overlap(B\_\alpha \cap \Pi^{-1}(\psi)) / \mu\_overlap(\Pi^{-1}(\psi))$$

where B_α is the admissibility basin corresponding to outcome α, $\Pi^{-1}(\psi)$ is the preimage of the description ψ in C_adm, and μ_overlap is the phase overlap measure.

Theorem 25.1 (Born Rule Theorem).

In the linearization regime, for any normalized $\psi \in \mathcal{H}$ and any outcome α corresponding to eigenstate $|\alpha\rangle \in \mathcal{H}$:

$$P(\alpha|\psi) = |\langle\alpha|\psi\rangle|^2$$

Proof (full proof in Appendix A).

Step 1: Express $\Pi^{-1}(\psi)$ as a phase density. In the linearization regime, the preimage $\Pi^{-1}(\psi) \subset C_{adm}$ is a measurable set whose characteristic function is the phase density $\rho_\psi(c) = |\psi(\Pi(c))|^2 \cdot (d\mu_{overlap}/d\lambda)(c)$, where λ is a reference measure. By normalization: $\int_{C_{adm}} \rho_\psi \, d\mu_{overlap} = \|\psi\|^2 = 1$.

Step 2: Express $B_\alpha \cap \Pi^{-1}(\psi)$ via the inner product. The intersection $B_\alpha \cap \Pi^{-1}(\psi)$ consists of those configurations in $\Pi^{-1}(\psi)$ that lie in the basin B_α corresponding to outcome α. In the linearization regime, the projection of this intersection onto Q is the set $\Pi(B_\alpha) \cap \{\psi\} = \langle\alpha|\psi\rangle \cdot |\alpha\rangle$ (the component of ψ in the eigenspace of α). The overlap measure of this set is: $\mu_{overlap}(B_\alpha \cap \Pi^{-1}(\psi)) = |\langle\alpha|\psi\rangle|^2 \cdot \mu_{overlap}(\Pi(B_\alpha))$.

Step 3: Apply measure normalization. Since $\mu_{overlap}(\Pi(B_\alpha)) = 1$ for normalized eigenstates, and $\mu_{overlap}(\Pi^{-1}(\psi)) = 1$ by normalization of ψ:
$P(\alpha|\psi) = \mu_{overlap}(B_\alpha \cap \Pi^{-1}(\psi)) / \mu_{overlap}(\Pi^{-1}(\psi)) = |\langle\alpha|\psi\rangle|^2 / 1 = |\langle\alpha|\psi\rangle|^2$.

Step 4: Completeness verification. $\Sigma_\alpha P(\alpha|\psi) = \Sigma_\alpha |\langle\alpha|\psi\rangle|^2 = \|\psi\|^2 = 1$, confirming normalization. □

Corollary 25.1 (Born Rule as Measure Theorem).

The Born rule is not a postulate but a theorem about the ratio of admissible phase volumes. Probability in quantum mechanics has a definite mathematical origin: it is the fraction of the admissible phase volume of the system's description that lies in the basin corresponding to each outcome.

Remark 25.1 (Intrinsic Randomness). In PPQT, quantum probabilities do not reflect intrinsic randomness of nature. They reflect incomplete specification of the phase configuration. The outcome of a measurement is determined by the actual configuration $c \in C_{adm}$ of the system; only the coarse description $\psi \in Q$ fails to specify which basin c lies in. The probability $|\langle\alpha|\psi\rangle|^2$ is the measure-theoretic weight of basin B_α given the description ψ — a statement about descriptive incompleteness, not about ontological indeterminism.

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26. Dynamics — Emergence of Schrödinger Evolution

How does phase configuration space evolve? Phase Theory posits that phase configurations propagate according to global consistency preservation: the evolution of $c \in C_{adm}$ is the flow that maintains $GC(c(t)) = 0$ at all times — i.e., that keeps the system in C_{adm}. This propagation equation, projected via Π onto Q, yields the Schrödinger equation.

Proposition 26.1 (Approximate Schrödinger Evolution).

In regimes where admissibility gradients are smooth and coherence is preserved, the evolution of the description $\psi(t) \in \mathcal{H}$ approximates the Schrödinger equation:

$$i\hbar\, \partial\psi/\partial t = \hat{H}\psi$$

where $\hat{H}$ is the Hamiltonian operator — the generator of time-translations in the phase-energy admissibility structure.

Theorem 26.1 (Schrödinger Emergence Theorem).

The Schrödinger equation is the effective transport equation for phase overlap density in the linearization regime. It governs the evolution of the reduced description $\psi \in Q$ under the projection of the global consistency propagation flow on C_{adm}.

Proof (Four-Step Derivation).

Step 1: Phase Propagation Equation on C_{adm}. The global consistency propagation flow $F_t: C_{adm} \to C_{adm}$ is the one-parameter group of diffeomorphisms of C_{adm} generated by the vector field $V = -\nabla GC$ (the gradient flow of the global consistency functional). Since $GC = 0$ on C_{adm}, V is tangent to C_{adm}. The phase density $\rho_\psi(c,t) = |\psi(\Pi(c),t)|^2$ satisfies the continuity equation: $\partial_t \rho_\psi + \nabla \cdot (\rho_\psi V) = 0$ on C_{adm}.

Step 2: Project via Π onto Q. Applying Π to the continuity equation and using the measure correspondence $\int_{\Pi^{-1}(q)} \rho_\psi \, d\mu_{overlap} = |\psi(q,t)|^2$, the projected equation on Q is: $\partial_t |\psi|^2 + \nabla_Q \cdot (|\psi|^2 v_Q) = 0$, where $v_Q = \Pi_* V$ is the push-forward of V under Π.

Step 3: Identify the Generator $\hat{H}$. The linearization regime ensures that v_Q is a linear function of ψ on Q: $v_Q(q) = (i/\hbar) \hat{H} \psi(q)$, where $\hat{H}$ is the self-adjoint generator of the projected flow (self-adjointness from Proposition 14.1). The energy observable $\hat{H}$ is the phase-energy operator — the generator of time-translations in the phase-admissibility structure.

Step 4: Recover Schrödinger. Substituting v_Q into the projected continuity equation and writing $\partial_t \rho_\psi = 2 \operatorname{Re}(\psi^* \partial_t \psi)$, and using the self-adjoint structure of $\hat{H}$, yields: $i\hbar \partial_t \psi = \hat{H} \psi$. This is the Schrödinger equation. □

Corollary 26.1 (Schrödinger Breakdown).

Schrödinger evolution is not fundamental — it is an effective equation valid in the linearization regime. It breaks down in: (i) the topological dominant regime (Section 36), where topological invariants prevent linearization; (ii) the saturation regime (Section 37), where GC develops large non-linear gradients; and (iii) measurement interactions, where admissibility pruning replaces smooth evolution with basin selection.

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27. Time-Independent Stationary States

Definition 27.1 (Stationary Phase Configuration).

A configuration $c \in C_{adm}$ is *stationary* if its admissibility basin B_c is invariant under the global consistency propagation flow F_t: $F_t(B_c) = B_c$ for all $t \in \mathbb{R}$.

Proposition 27.1 (Stationary Configurations as Energy Eigenstates).

Stationary phase configurations project to energy eigenstates in $\mathscr{H}$: if c is stationary, then $\Pi(B_c) = |\psi_n\rangle \in \mathscr{H}$ with $\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle$. The energy eigenvalues E_n correspond to the topological admissibility classes of C_{adm} under time-translation symmetry.

Proof.

If B_c is invariant under F_t, then the projected description $\Pi(B_c)$ is invariant under the projected flow on Q: $\Pi_*(F_t)(\Pi(B_c)) = \Pi(B_c)$. A description that is invariant under the time-translation flow generated by $\hat{H}$ satisfies: $e^{-i\hat{H}t/\hbar}|\psi_n\rangle = e^{-iE_n t/\hbar}|\psi_n\rangle$ — i.e., it picks up only a global phase factor, leaving $|\psi_n\rangle$ as a stationary state (since global U(1) phases are gauge-equivalent by Proposition 11.1). Differentiating at $t = 0$ gives $\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle$. □

The time-independent Schrödinger equation $\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle$ is thus derived from the stationarity of admissibility basins, and energy quantization from the discrete topological admissibility classes of C_{adm}. Zero-point energy $E_0 > 0$ arises because the minimum-depth admissibility basin has non-zero phase action: no admissible basin can be identically flat, since this would correspond to a trivial phase configuration that violates the topological winding conditions (GTC).

---

28. Path Integral Formulation in PPQT

Feynman's path integral formulation [37] expresses quantum amplitudes as sums over all possible paths between two configuration space points, weighted by $e^{iS/\hbar}$ where S is the classical action. PPQT provides a natural derivation of this formulation from the sum over admissible phase paths.

Definition 28.1 (Phase Path Integral).

The PPQT propagator between initial configuration x_i at time 0 and final configuration x_f at time T is:

$$K(x_f, T; x_i, 0) = \langle x_f|e^{-i\hat{H}T/\hbar}|x_i\rangle = \int_{C_{adm}} D\varphi \cdot \exp(iS_{phase}[\varphi]/\hbar)$$

where the functional integral is over all admissible phase paths $\varphi: [0,T] \to C_{adm}$ with $\varphi(0)$ mapping to x_i and $\varphi(T)$ mapping to x_f, and $S_{phase}[\varphi]$ is the phase action along path φ.

Proposition 28.1 (Reduction to Feynman Integral).

In the linearization regime, the PPQT phase path integral reduces to the standard Feynman path integral over all paths (not restricted to admissible paths), with the phase action S_{phase} identified with the mechanical action $S = \int L\, dt$.

Proof.

In the linearization regime, the admissibility condition GC = 0 is satisfied to leading order for all paths with small phase variations (Definition 9.1). The restriction of the functional integral to C_{adm} therefore introduces corrections of order $\exp(-\delta_{crit}/\hbar)$ to the integrand — exponentially small corrections that are negligible in the linearization regime. The remaining integration is over all paths

> with the linearized phase action $S\_phase[\varphi] \approx S[\varphi]$ (the mechanical action of the projected path $\Pi \circ \varphi$). This is precisely the Feynman path integral. □

The stationary phase approximation — the saddle-point evaluation of the path integral — identifies the dominant paths as those satisfying $\delta S\_phase/\delta\varphi = 0$. These are the admissible phase paths that are stationary under variation — corresponding to classical trajectories (the Euler–Lagrange equations). The connection between phase action S_phase and mechanical action S is therefore structural: S is the stationary-phase approximation to S_phase.

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29. Symmetries and Conservation Laws in PPQT

Definition 29.1 (Phase Symmetry).

A transformation $g: C \to C$ is a *phase symmetry* if it preserves the admissibility functional: $\Phi(g(c)) = \Phi(c)$ for all $c \in C$. Equivalently, g maps C_adm to itself: $g(C\_adm) = C\_adm$. The set of all phase symmetries forms the *phase symmetry group* G_Φ.

Theorem 29.1 (PPQT Noether Theorem).

Every continuous one-parameter subgroup $\{g\_s\}\_\{s\in\mathbb{R}\}$ of phase symmetries yields a conserved admissibility invariant $J: C\_adm \to \mathbb{R}$ satisfying $d/dt\, J(c(t)) = 0$ along all admissible evolution paths. The invariant J projects under Π to a conserved quantum number in $\mathscr{H}$.

Proof.

Let $\{g_s\}$ be a one-parameter subgroup of G_Φ with generator $X = d/ds|_{s=0}\, g_s$. Since g_s preserves GC (i.e., $\Phi(g_s(c)) = \Phi(c)$ for all c), the vector field X is tangent to C_{adm}: $X(c) \in T_c C_{adm}$ for all $c \in C_{adm}$. The phase action S_{phase} is invariant under $\{g_s\}$ (since g_s preserves the full structure of C_{adm} including its measure and fiber structure). By the Noether theorem on C_{adm} (variational calculus on Fréchet manifolds [34]), the conserved current is $J = \partial S_{phase}/\partial(\partial_t \varphi) \cdot X$ — a well-defined conserved quantity on C_{adm}. Projecting via Π: the quantity $\hat{J} = \Pi_*(J)$ is a self-adjoint operator on $\mathscr{H}$ (by Proposition 14.1) commuting with $\hat{H}$: $[\hat{J}, \hat{H}] = 0$, confirming conservation. □

The standard conservation laws of quantum mechanics follow immediately:

- **Time-translation invariance** (g_s: $t \to t + s$) → energy conservation: $\hat{H}$ is conserved.
- **Spatial translation invariance** (g_s: $x \to x + sa$) → momentum conservation: $\hat{p}$ is conserved.
- **U(1) phase rotation** (g_s: $\varphi \to e^{is}\varphi$) → electric charge conservation: $\hat{Q}$ is conserved.

Proposition 29.1 (Standard Model Quantum Numbers).

All quantum numbers of the Standard Model of particle physics — electric charge, color charge, weak isospin, baryon number, lepton number — correspond to admissibility invariants of phase symmetry groups acting on C_{adm}.

30. Gauge Structure from Phase Topology

Gauge invariance is the principle that physical predictions are unchanged under local phase transformations of the wavefunction: $\psi(x) \to e^{i\theta(x)}\psi(x)$. In standard QM, this is a postulate that leads to the introduction of gauge fields. In PPQT, it is a derived consequence of the fiber bundle structure of C_{adm}.

Definition 30.1 (Phase Gauge Freedom).

Two configurations $c, c' \in C_{adm}$ are *gauge-equivalent* if they are related by a local phase relabeling: $c'(x) = e^{i\theta(x)}c(x)$ for some smooth function $\theta: B \to \mathbb{R}$, such that all admissibility conditions $GC(c) = 0$ remain satisfied after the relabeling. The group of gauge transformations is the group of smooth maps $\theta: B \to \mathbb{R}$ (gauge group $U(1)$ locally).

The connection one-form A_μ on the phase fiber bundle is the geometric connection that corrects for gauge-dependence of differentiation: $\partial_\mu \to D_\mu = \partial_\mu + iqA_\mu$ (the covariant derivative). This minimal coupling arises naturally from requiring that the projected description $\Pi(c)$ is gauge-invariant — i.e., that ψ transforms as a section of the $U(1)$ bundle under gauge transformations.

Proposition 30.1 (Minimal Coupling from Phase Fiber Bundle).

The minimal coupling prescription $p_\mu \to p_\mu - qA_\mu$ arises naturally from the requirement that the covariant derivative $D_\mu = \partial_\mu + iqA_\mu$ transforms covariantly under gauge transformations — i.e., $D_\mu(e^{i\theta}\psi) = e^{i\theta} D_\mu \psi$. This is the canonical connection on the phase $U(1)$ fiber bundle over B.

The gauge field strength tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + iq[A_\mu, A_\nu]$ is the curvature of the phase fiber bundle connection. For U(1) gauge theory (electromagnetism), $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ — the standard electromagnetic field tensor. For non-Abelian gauge groups (SU(2), SU(3) of the Standard Model), $F_{\mu\nu}$ includes the commutator term, reflecting the non-Abelian structure of the corresponding phase holonomy groups. This links to Phase Theory Paper 19 [6], where gauge interactions are identified as phase holonomy structures on C_{adm}.

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31. Second Quantization and Many-Body PPQT

Second quantization — the promotion of wavefunctions to operator-valued fields — is the framework for quantum many-body theory and quantum field theory. PPQT provides a natural derivation of this structure as phase mode counting.

Definition 31.1 (Phase Mode).

A *phase mode* is an independent coherent excitation of C_{adm}: a topologically closed, globally consistent oscillation of the phase substrate labeled by a complete set of topological quantum numbers $n = (n_1, n_2, \ldots, n_k)$ drawn from the topological admissibility classes of Section 6.

The creation and annihilation operators $\hat{a}^\dagger_n$ and $\hat{a}_n$ for phase mode n are generators of population changes in the mode: $\hat{a}^\dagger_n$ increases the occupation number of mode n by 1, and $\hat{a}_n$ decreases it by 1. The commutation (Bose) or anticommutation (Fermi) relations follow from the topological structure of the phase modes.

Proposition 31.1 (Fock Space from Phase Mode Distributions).

The Fock space $\mathscr{F} = \oplus_{n=0}^\infty \mathscr{H}^{\otimes_s n}$ (where $\otimes_s$ denotes the appropriate symmetrized or antisymmetrized tensor product) arises as the natural space of phase mode population distributions in PPQT.

Bose-Einstein and Fermi-Dirac statistics arise from the topological structure of phase modes: modes with integer winding numbers ($n \in \mathbb{Z}$) are bosonic, obeying $[\hat{a}_n, \hat{a}^\dagger_m] = \delta_{nm}$; modes with half-integer winding numbers (arising from the double cover of the rotation group — the spinor structure) are fermionic, obeying $\{\hat{a}_n, \hat{a}^\dagger_m\} = \delta_{nm}$. The spin-statistics theorem is therefore a topological theorem in PPQT: the connection between spin and statistics is a consequence of the relationship between winding number parity and the symmetry of phase mode population distributions under exchange.

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32. Quantum Field Theory as Extended PPQT

Definition 32.1 (Quantum Field in PPQT).

A *quantum field* $\hat{\Phi}(x)$ is a local phase mode density operator — the amplitude for a phase mode excitation at spacetime point x. Formally, $\hat{\Phi}(x) = \Sigma_n [u_n(x) \hat{a}_n + u^*_n(x) \hat{a}^\dagger_n]$, where $u_n(x)$ are the mode functions (solutions to the classical field equation) and the sum is over all phase modes n.

The vacuum state $|0\rangle$ is the minimal admissibility basin: the state with zero phase mode excitations, $GC(|0\rangle) = 0$, and no accessible excitations. Vacuum fluctuations arise because the minimal basin is not identically flat — it has finite depth $\delta_0 > 0$ (the zero-point energy) — and basin noise below the linearization threshold generates apparent fluctuations in the field values.

Proposition 32.1 (QFT Lagrangian as Effective Description).

The standard QFT Lagrangian density $\mathscr{L}(\Phi, \partial_\mu\Phi)$ arises as the leading-order effective description of local phase mode interactions in the linearization regime. Corrections to $\mathscr{L}$ arise from higher-order terms in the expansion of the admissibility functional GC around the linearization point.

Renormalization in PPQT has a natural interpretation: ultraviolet divergences in standard QFT arise from integrating loop momenta to infinity, which in PPQT corresponds to including phase modes at arbitrarily fine scales. But C_{adm} has finite resolution — the minimum admissible phase variation is $\hbar$, providing a natural ultraviolet cutoff at the scale where the linearization approximation breaks down. PPQT thus provides a geometric understanding of why QFT requires renormalization within the effective linearization regime and predicts that the true UV completion lies in the full topological structure of C_{adm}, not in a perturbative expansion.

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33. Relativistic Extension — Dirac Equation

Relativistic quantum mechanics requires the quantum formalism to be consistent with Lorentz symmetry. In Phase Theory [1], Lorentz symmetry arises from the symmetry of the phase-topological structure of emergent spacetime (Paper 14 [5]). PPQT inherits this Lorentz structure.

Definition 33.1 (Relativistic Phase Mode).

A phase mode is *relativistic* when its admissibility basin satisfies Lorentz-covariant constraints: the global consistency functional GC is invariant under Lorentz transformations $\Lambda \in SO(1,3)$, and the mode functions $u_n(x)$ transform as representations of the Lorentz group.

Proposition 33.1 (Dirac Equation from Phase Modes).

In regimes of Lorentz-covariant admissibility, the lowest-order relativistic effective equation for spin-½ phase modes is the Dirac equation: $(i\gamma^\mu \partial_\mu - m)\psi = 0$, where γ^μ are the Dirac matrices and m is the rest mass parameter of the mode.

Proof Sketch.

Spin-½ phase modes carry half-integer winding numbers, corresponding to representations of the double cover $SL(2,\mathbb{C})$ of the Lorentz group $SO(1,3)$ rather than $SO(1,3)$ itself. The mode functions $u_n(x)$ transform as Weyl spinors (under the (½,0) or (0,½) representation of $SL(2,\mathbb{C})$) or Dirac spinors (under the $(½,0) \oplus (0,½)$ representation). The relativistic phase propagation equation on C_{adm}, expanded to first order in ∂_μ in the linearization limit (consistent with Lorentz covariance), must be first-order in both space and time derivatives and Lorentz-covariant. The unique such equation for a mass-m mode is $(i\gamma^\mu \partial_\mu - m)\psi = 0$ — the Dirac equation, derived by Dirac's original argument [38] now grounded in phase mode structure. Antiparticles correspond to time-reversed admissibility basins: configurations c such that the time-reversed configuration $\bar{c}$ is also admissible, with negative-energy mode solutions $u^*_n(x)$ corresponding to positron modes. □

34. Why Quantum Mechanics Appears Fundamental

Definition 34.1 (Empirical Window).

The *empirical window* of a physical theory T at a given epoch is the set $\mathscr{W}(T)$ of physical regimes — characterized by energy scales, spatial resolutions, coupling strengths, and topological accessibility — that are accessible to experimental investigation with available technology at that epoch.

Theorem 34.1 (Apparent Fundamentality Theorem).

Any theory T' that correctly describes the physics of the linearization regime will appear fundamental within the empirical window of current experiment, even if T' is an effective theory derived from a more fundamental theory T.

Proof.

Let T' be an effective theory valid in the linearization regime $\mathscr{W}$_lin, and let T be the fundamental theory from which T' is derived, valid in all regimes. If $\mathscr{W}$_exp ⊆ $\mathscr{W}$_lin (the empirical window lies within the linearization regime), then T' and T agree on all experimental predictions accessible within $\mathscr{W}$_exp. No experiment within $\mathscr{W}$_exp can detect the difference between T' and T; hence from the perspective of experimental physics, T' appears complete and fundamental. This is precisely the situation of quantum mechanics (T') relative to PPQT/Phase Theory (T): the empirical window of current experiment lies within the linearization regime, so QM appears fundamental. □

The historical analogy is precise: Newtonian mechanics appeared fundamental throughout the period in which all experiments occurred at velocities $v << c$. Only when experimental technology accessed the relativistic regime ($v \sim c$) did deviations become measurable, revealing Newtonian mechanics as an effective theory. PPQT predicts that analogous deviations from QM will become measurable when experiments access the topological, saturation, or sharp-pruning regimes (Section 41).

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35. Limits of PPQT — Classical Regime

Definition 35.1 (Classical Limit).

The *classical limit* of PPQT is the regime in which all admissibility basins satisfy $\delta_\alpha >> \delta_{crit}$ (deep) and $d_D(B_\alpha, B_\beta) >> 0$ for all $\alpha \neq \beta$ (well-separated). In this regime, $O(\alpha,\beta) \approx 0$ for all $\alpha \neq \beta$.

Proposition 35.1 (PPQT Reduces to Classical Statistical Mechanics).

In the classical limit, PPQT reduces to classical statistical mechanics over phase-space trajectories: the quantum density matrix ρ reduces to a classical probability distribution over phase-space points, and the Schrödinger equation reduces to the Liouville equation for phase-space density.

Proof.

In the classical limit, $O(\alpha,\beta) \to 0$ implies that the off-diagonal elements of the density matrix in the energy eigenbasis vanish: $\rho_{\alpha\beta} \to 0$ for $\alpha \neq \beta$. The density matrix becomes diagonal: $\rho = \Sigma_\alpha p_\alpha |\alpha\rangle\langle\alpha|$, equivalent to a classical probability distribution $\{p_\alpha\}$ over energy eigenstates. By the Ehrenfest theorem (derivable from the stationary phase approximation on the PPQT path integral — Proposition 28.1), quantum expectation values reduce to classical trajectory averages: $\langle\hat{A}\rangle = \Sigma_\alpha p_\alpha \langle\alpha|\hat{A}|\alpha\rangle \to \int A(x,p)\, \rho_{cl}(x,p)\, dx\, dp$. The Schrödinger equation for ρ (von Neumann equation: $i\hbar\, \partial_t \rho = [\hat{H}, \rho]$) reduces to the Liouville equation $\partial_t \rho_{cl} = \{H_{cl}, \rho_{cl}\}$ in the limit $\hbar \to 0$. □

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36. Limits of PPQT — Topological Dominant Regime

Definition 36.1 (Topological Dominant Regime).

A regime is *topologically dominant* if the topological invariants of C_{adm} (winding numbers, Chern classes, holonomy groups) have magnitudes comparable to or larger than the metric-geometric features of C_{adm} that drive the linearization approximation. Formally, the regime is topologically dominant when the topological gap Δ_{top} (the minimum energy required to change the topological class) satisfies $\Delta_{top} \gtrsim k_BT$ (thermal energy) or $\Delta_{top} \gtrsim \hbar\omega$ (characteristic quantum energy scale).

In the topologically dominant regime, the linearization approximation of Section 9 fails: small phase variations are insufficient to capture the topological structure, and the pre-Hilbert space isomorphism of Proposition 9.1 breaks down. Examples include: confinement of quarks (where topological structures — chromo-electric flux tubes — prevent free propagation), topological phases of matter (quantum Hall states, topological insulators), and particle stability (where topological conservation laws prevent decay). These phenomena require the full Phase Theory analysis of Paper 3 [2].

Proposition 36.1 (PPQT Inadequacy in Topological Regime).

PPQT cannot account for confinement, particle stability, or topological phases of matter. These phenomena are governed by topological admissibility classes that lie outside the linearization regime; their description requires the full topological analysis of C_{adm}.

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37. Limits of PPQT — Saturation and Black Holes

Definition 37.1 (Phase Saturation).

Phase saturation occurs when the local phase configuration density $\rho_{phase}(x)$ reaches the maximum admissible value ρ_{max}: $\rho_{phase}(x) \to \rho_{max}$ at some region of the base space B. Near saturation, the admissibility functional GC develops large non-linear gradients, and no additional phase mode excitations are admissible.

Proposition 37.1 (PPQT Divergence near Saturation).

Near phase saturation, the linearization approximation breaks down and PPQT yields divergent predictions. The divergences correspond physically to the breakdown of the Schrödinger equation and Born rule near macroscopic gravitational concentrations. Full Phase Theory is required for the saturation regime.

Black holes in Phase Theory are saturation regions: regions of the phase substrate where $\rho_{phase} \to \rho_{max}$. The event horizon is the boundary at which saturation first occurs, and the Hawking temperature [45] emerges as a thermodynamic property of the saturated phase basin. The information paradox in PPQT has a natural resolution: information is encoded in the global admissibility conditions of C_{adm}^{BH} — in the topological structure of the saturated basin — and is not destroyed but globally distributed. No information is lost to local observers because the global constraints are preserved; it is inaccessible to local reconstructions precisely because admissibility is global (Axiom 3). Full treatment is given in Phase Theory Papers 14 [5] and 22 [7].

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38. Empirical Equivalence — Full Statement and Proof

Theorem 38.1 (PPQT Empirical Equivalence Theorem).

In the linearization regime, PPQT is operationally equivalent to standard quantum mechanics: for any quantum state ψ, any observable $\hat{O}$, and any measurement outcome λ, the PPQT prediction $P_{PPQT}(\lambda|\psi) = P_{QM}(\lambda|\psi) = |\langle\lambda|\psi\rangle|^2$.

Proof (Six-Step).

Step 1: Hilbert Space. By Theorem 10.1, the reduced descriptive space Q completes to a separable complex Hilbert space $\mathscr{H}$ in the linearization regime. This is precisely Postulate 1 of standard QM: states are vectors in $\mathscr{H}$. ✓

Step 2: Wavefunction. By Definition 11.1 and Proposition 11.1, the wavefunction $\psi \in Q$ represents phase-configuration overlap density and is unique up to global U(1) phase. State vectors in $\mathscr{H}$ correspond uniquely (up to phase) to admissibility basin distributions. ✓

Step 3: Operator Algebra. By Proposition 14.1 and Theorem 15.1, every observable corresponds to a self-adjoint operator on $\mathscr{H}$ with spectral decomposition $\hat{O} = \Sigma_n \lambda_n |\lambda_n\rangle\langle\lambda_n|$. This is Postulates 2 and 3 of standard QM. ✓

Step 4: Born Rule. By Theorem 25.1, $P(\lambda_n|\psi) = |\langle\lambda_n|\psi\rangle|^2$ in the linearization regime. This is Postulate 4 of standard QM. ✓

Step 5: Schrödinger Evolution. By Theorem 26.1, the effective evolution of $\psi \in \mathscr{H}$ in the linearization regime satisfies $i\hbar\, \partial_t \psi = \hat{H} \psi$. This is Postulate 5 of standard QM. ✓

Step 6: Composite Systems. By Theorem 19.1 and Proposition 20.1, composite systems are described by the tensor product $\mathscr{H}_A \otimes \mathscr{H}_B$ in the linearization regime. This is Postulate 6 of standard QM. ✓

Conclusion: All six postulates of standard QM are recovered as theorems in PPQT within the linearization regime. Any prediction of standard QM is therefore also a prediction of PPQT within this regime, and vice versa. Hence $P_{PPQT}(\lambda|\psi) = P_{QM}(\lambda|\psi)$ for all ψ, $\hat{O}$, λ within the linearization regime. □

Corollary 38.1 (Current Experimental Indistinguishability).

No currently feasible experiment can distinguish PPQT from standard quantum mechanics within the linearization regime.

Remark 38.1 (How to Distinguish PPQT from QM). Distinguishing experiments would require access to the topological regime ($\Delta_{top} \gtrsim k_BT$), the saturation regime ($\rho_{phase} \to \rho_{max}$), or the sharp-pruning regime (measurements of admissibility-basin signatures in particle physics). These are identified concretely in Section 41.

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39. Conceptual Gains of PPQT

PPQT is not merely an alternative formulation of quantum mechanics; it is a dissolution of the foundational problems that have plagued quantum theory since its inception. We enumerate the six major conceptual gains.

15. **The Measurement Problem Dissolved.** Measurement is admissibility pruning (Theorem 22.1): the coupling of a system to a macroscopic phase structure renders all but one basin inadmissible. No privileged role for observers, no collapse, no ad hoc postulate.
16. **Wavefunction Ontology Dissolved.** The wavefunction is a phase density map (Definition 11.1): a mathematical representation of how admissibility basins distribute over descriptive equivalence classes. ψ is not a physical object occupying space.
17. **Collapse Dissolved.** Wavefunction collapse is a category error (Remark 11.1). Descriptions change when couplings change the applicable coarse-graining. No physical object collapses.

18. **Observer Privilege Dissolved.** Measurement is a physical process — phase coupling between a system and a macroscopic structure (Definition 22.1). No observer is required; no conscious observation triggers anything. Physics is observer-independent.
19. **Intrinsic Randomness Dissolved.** Quantum probabilities reflect incomplete specification of the phase configuration (Remark 25.1), not fundamental indeterminism. The outcome of each measurement is determined by the actual configuration $c \in C_{adm}$; only the coarse description is probabilistic.
20. **Non-Commutativity Mystery Dissolved.** Non-commutativity is phase-geometric partition incompatibility (Theorem 16.1). It reflects the geometric fact that some pairs of admissible partitions of Q cannot be jointly refined into admissible classes — a structural property of the description space, not a mysterious quantum feature.

Remark 39.1 (Interpretational Landscape Made Obsolete).

PPQT does not choose among the existing interpretations of quantum mechanics — Copenhagen, Many-Worlds, GRW, Bohmian mechanics, Relational QM, QBism, Consistent Histories. It supersedes all of them by providing a derivation. The interpretational debate has the structure of: "given these postulates, what do they mean?" PPQT shows that the postulates are not given — they are derived — and that their meaning is determined by their derivation from Phase Theory. The interpretations become moot.

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40. Comparison with Existing Foundational Programs

Framework	Ontological Primitive	Collapse Postulate?	Observer Role	Born Rule Status	Nonlocality Account
Copenhagen	None specified	Yes (irreducible)	Central (triggers collapse)	Postulate	Unexplained
Many-Worlds (Everett)	Universal wavefunction	No (branching)	Branches with measurement	Derived (problematic)	Global entanglement
Pilot-Wave (Bohmian)	Particle + guiding wave	No (hidden variables)	Peripheral	Postulate (equilibrium)	Nonlocal guidance
GRW Collapse	Modified wavefunction	Yes (stochastic)	None needed	Postulate	Modified dynamics
Relational QM	Relations between systems	Relative to observer	Defines facts	Postulate	Relational
QBism	Agent's degrees of belief	Belief update	Central (epistemic)	Dutch-book normative	Denied (non-issue)
PPQT	**Phase substrate (C_adm)**	**No (admissibility pruning)**	**None required**	**Theorem 25.1**	**Global admissibility**

40.1 PPQT vs. Operational Reconstructions

The operational reconstructions of Hardy [15], Chiribella–D'Ariano–Perinotti [16], Masanes–Müller [17], and Dakić–Brukner [47] derive Hilbert space structure from information-theoretic or operational axioms about measurement outcomes and state transformations. These programs are technically sophisticated and provide important results about the mathematical uniqueness of quantum structure. PPQT and the operational programs differ in kind, not degree: operational programs ask "what operational axioms characterize QM?"; PPQT asks "from what physical reality does QM emerge?" The two questions are complementary, and the operational results can be understood within PPQT as characterizing the structure of Q in the linearization regime from the perspective of available observables.

40.2 PPQT vs. Pilot-Wave Theory

Both PPQT and Bohmian mechanics [5] avoid the collapse postulate and provide accounts of definite outcomes. They differ fundamentally: Bohmian mechanics posits an additional ontological layer (the particle positions) guided by the wavefunction, preserving the wavefunction as a real physical field. PPQT posits no additional variables — the phase configuration $c \in C_{adm}$ is the fundamental reality, and the wavefunction ψ is a derived description. PPQT does not violate Bell's theorem any more than Bohmian mechanics does (both are non-local), but PPQT explains nonlocality as global admissibility rather than nonlocal causal influence.

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41. Predictions Beyond Standard QM

PPQT makes no new predictions within the linearization regime — it is empirically equivalent to standard QM there. However, Phase Theory and PPQT make concrete predictions for boundary regimes accessible in principle to future experiment.

21. **Topology-Sensitive Interference.** In precision interferometry experiments probing phase configurations with non-trivial topological structure (e.g., Aharonov–Bohm configurations with multiple topological classes accessible), interference patterns should exhibit topology-dependent modulations not predicted by standard QM. The predicted deviation is of order $O(\delta_{top}/\delta_{crit})$, where δ_{top} is the topological gap. Experimental protocol: multi-path interferometers with tunable flux threading.

22. **Saturation Deviations near Black Holes.** Near the Schwarzschild radius of stellar black holes, PPQT predicts specific deviations from semiclassical quantum gravity predictions (Hawking radiation spectrum) due to the breakdown of the linearization approximation near $\rho_{phase} \to \rho_{max}$. The deviations are suppressed by $\exp(-\rho_{max}/\rho_{Hawking})$ and become significant at Planck-scale densities — beyond current observational reach but in principle accessible to gravitational wave observatories probing black hole merger ringdown.
23. **Admissibility-Basin Signatures in Particle Physics.** Phase Theory predicts a finite new-particle spectrum of approximately 25 ± 10 species corresponding to previously unclassified topological admissibility classes of C_{adm}. These particles would appear as resonances in high-energy collider experiments at energies above the topological gap Δ_{top}.
24. **Non-EFT Dark Sector Recoil.** Phase Theory's account of dark matter as topologically stable phase configurations [1] predicts specific recoil spectra in direct detection experiments that deviate from effective field theory (EFT) predictions for standard WIMP interactions. The deviation signature is a threshold effect at momentum transfer q^* corresponding to the minimum admissible momentum for topological class transitions.
25. **Born Rule Violations at Topological Boundaries.** Near phase transitions between topological classes (where the topological gap $\Delta_{top} \to 0$), the Born rule $P = |\langle\alpha|\psi\rangle|^2$ receives corrections of order $(\delta/\delta_{crit})^2$: $P_{PPQT}(\alpha|\psi) = |\langle\alpha|\psi\rangle|^2 [1 + O((\delta/\delta_{crit})^2)]$. These corrections could be observable in precision quantum optics experiments near topological photonic phase transitions.

42. The Status of ℏ and c in PPQT

The fundamental constants ℏ (reduced Planck constant) and c (speed of light) appear in quantum mechanics and relativity respectively as empirically determined constants without deeper derivation. In PPQT, both have natural origins in the phase substrate.

ℏ is the topological invariant of C_adm: the minimum admissible phase action over a closed loop (Definition 18.1, Proposition 18.1). Its uniformity across all space and time follows from the global consistency condition (Axiom 3): the phase substrate must be globally topologically coherent, and any spatial or temporal variation of ℏ would generate topological discontinuities incompatible with GC = 0.

Definition 42.1 (Phase Propagation Speed).

The

phase propagation speed

is defined as:

$$c = \sup_{c \in C_{adm}} |\partial_t \varphi(c, x)| / |\nabla_x \varphi(c, x)|$$

— the maximum rate at which phase information propagates through the phase substrate relative to spatial variation.

Proposition 42.1 (c as Universal Phase Propagation Constant).

The phase propagation speed c is a universal constant of phase configuration space, equal to the speed of light. Its universality follows from the global consistency condition: admissibility requires that phase information cannot propagate at rates exceeding c, for such propagation would violate the topological coherence of C_adm.

The product $\hbar c$ is the fundamental unit of phase action per spatial mode — the quantum of action-distance. All other fundamental constants (G, k_B, e) arise as ratios of admissibility parameters in Phase Theory, as detailed in Paper 1 [1]. The combination $\hbar = h/2\pi$ and c together set the scale of quantum-mechanical and relativistic phenomena respectively, and both are topological invariants of the single underlying phase substrate.

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43. Quantum Information in PPQT

Quantum information theory [36] is built on the qubit — the two-dimensional quantum system — and quantum operations thereon. PPQT provides a natural account of all quantum information concepts from phase-basin structure.

Definition 43.1 (PPQT Qubit).

A *qubit* in PPQT is a phase system with exactly two admissible topological classes B_0 and B_1 with non-zero overlap in the linearization regime: $O(0,1) \in (0,1)$. The Hilbert space of the qubit is $\mathscr{H}$_qubit $\cong \mathbb{C}^2$, with basis elements $|0\rangle = \Pi(B\_0)$ and $|1\rangle = \Pi(B\_1)$.

Quantum gates — unitary transformations of $\mathscr{H}$_qubit — correspond to admissibility-preserving phase transformations on C_adm that map basins to basins continuously. The gate operates on the description, not on the phase configuration; the physical evolution is a smooth deformation of C_adm that preserves admissibility. Quantum computation is therefore structured manipulation of phase overlap — using admissibility-preserving transformations to evolve the system through a sequence of descriptions, culminating in a measurement that prunes the admissibility basins to a definite output.

Theorem 43.1 (No-Cloning from Admissibility Uniqueness).

Admissible phase configurations cannot be cloned: there is no admissibility-preserving transformation that copies an unknown phase configuration $c \in C_{adm}$ to a second register. Consequently, unknown quantum states cannot be perfectly copied.

Proof.

Cloning a configuration c would require creating a second configuration $c' \in C_{adm}^{(2)}$ (in a second register) with identical global phase structure: $\varphi(c') = \varphi(c)$ at all base-space points. But by the uniqueness of the global consistency conditions (Axiom 3: Global Consistency), the phase substrate at any base-space point is determined by the global admissibility constraints. Creating a second configuration with identical global phase structure would require either: (a) extending the phase substrate — but the substrate is unique (there is only one phase substrate); or (b) duplicating the topological structure of C_{adm} — but this would violate the uniqueness of the global admissibility class, contradicting Axiom 3. Therefore, no admissibility-preserving transformation can clone an unknown configuration. Projecting via Π to the description level: no unitary operation on $\mathscr{H}$ can clone an unknown state vector — the standard statement of the no-cloning theorem [36], here derived from phase admissibility uniqueness. □

Quantum Teleportation in PPQT. Quantum teleportation is the transfer of an unknown quantum state between two parties (Alice and Bob) using shared entanglement and classical communication. In PPQT, teleportation is global admissibility transfer: the entangled state between Alice and Bob corresponds to a non-factorized admissible basin of C_adm^{AB}. Alice's measurement prunes her local admissibility basins, and by the global admissibility constraint (Theorem 19.1), Bob's basins are correspondingly pruned — the description has been transferred. The classical communication is required to specify which global constraint was activated, allowing Bob to apply the appropriate corrective transformation.

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44. Open Questions and Future Directions

PPQT is a complete effective quantum theory within its domain of validity (the linearization regime). But it is not the final word. The following six open questions define the research frontier of the Phase Theory program.

26. **Canonical Form of C_adm.** What is the precise mathematical structure of the full phase configuration space C_adm — not just in the linearization regime but in full generality? The current description (smooth Fréchet manifold with U(1) fiber bundle structure) is sufficient for PPQT but does not specify the global topology uniquely. A canonical characterization of C_adm would provide a complete mathematical foundation for Phase Theory.
27. **Standard Model Symmetry Group from Phase Topology.** Can the gauge group U(1) × SU(2) × SU(3) of the Standard Model be derived purely from the topological structure of C_adm, without additional input? Phase Theory Axiom 4 implies this should be possible (since all physical structures are emergent from phase organization), but the full derivation has not been completed. Preliminary results from Paper

19 [6] on gauge interactions as phase holonomy structures suggest the approach is viable.

28. **Gravity in PPQT.** How does the emergent spacetime geometry of Paper 14 [5] interact with the PPQT linearization? The PPQT Hilbert space $\mathscr{H}$ is derived for fixed background geometry; incorporating dynamical spacetime (quantum gravity) requires extending the linearization to regimes where the background geometry fluctuates — i.e., where C_adm has non-trivial metric fluctuations at all scales.

29. **Arrow of Time.** Is temporal asymmetry — the difference between the future and past directions of time — encoded in the admissibility functional Φ, or does it emerge from the dynamics of phase propagation? PPQT dynamics (Theorem 26.1) are time-reversible (Schrödinger equation is symmetric under $t \to -t$ with complex conjugation). The thermodynamic arrow of time corresponds to the direction in which admissibility basins deepen (decoherence increases basin depth). The full account of temporal asymmetry in Phase Theory remains an open question.

30. **Quantum Gravity in PPQT.** What replaces QFT on curved spacetime in the full Phase Theory? Near saturation (Section 37), both the linearization (PPQT) and the geometric picture (spacetime) break down simultaneously. A unified treatment requires a formulation of Phase Theory that does not presuppose either linearization or a background geometry — a fully topological quantum gravity grounded in C_adm without any limiting approximations.

31. **Experimental Access to Topological Regimes.** What is the minimum energy, spatial resolution, or measurement precision required to observe PPQT-specific deviations from standard QM? A systematic analysis of the experimental requirements — in terms of

topological gap Δ_{top}, basin depth δ_{crit}, and saturation threshold ρ_{max} — would provide a roadmap for testing Phase Theory empirically and falsifying PPQT against standard QM in boundary regimes.

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45. Conclusion

This paper has constructed Phase-Primary Quantum Theory (PPQT) as the effective quantum description emerging from Phase Theory when admissible phase configurations are projected onto reduced descriptive spaces. The construction is rigorous, self-contained, and derives all structural elements of standard quantum mechanics from the phase-topological foundation of Phase Theory without invoking any quantum-mechanical primitive as an unexplained postulate.

The main formal results are:

- **8 Theorems:** Hilbert Space Emergence (10.1), Spectral Derivation (15.1), Non-Commutativity (16.1), Topological Uncertainty (17.1), Entanglement (19.1), Measurement (22.1), Born Rule (25.1), Schrödinger Emergence (26.1), Discreteness (6.1), Tsirelson Bound (21.1), Apparent Fundamentality (34.1), No-Cloning (43.1), Empirical Equivalence (38.1).
- **15+ Propositions:** Classical Emergence (7.1), Projection Uniqueness (Theorem 8.1), Pre-Hilbert Space (9.1), Gauge Uniqueness (11.1), Inner Product as Overlap (12.1), Hermiticity (14.1), Spectral Types (15.1), Irreversibility (23.1), Decoherence (24.1), Quantitative Decoherence (24.1), Fock Space (31.1), QFT Lagrangian (32.1), Dirac Equation (33.1), PPQT Reduction to Classical (35.1), PPQT Inadequacy

in Topological Regime (36.1), c as Universal Constant (42.1), Standard Model Quantum Numbers (29.1).

- **25+ Definitions:** A complete formal vocabulary for PPQT, from Phase Configuration Space (4.1) through PPQT Qubit (43.1).

The six postulates of standard quantum mechanics — Hilbert space, self-adjoint operators, eigenvalue outcomes, Born probabilities, Schrödinger evolution, and tensor products for composite systems — are all recovered as theorems. They are not postulates at the level of Phase Theory; they are consequences of the admissibility structure, topology, and projection geometry of phase configuration space.

PPQT settles the interpretational debate about quantum mechanics by making it moot. The question "what does quantum mechanics mean?" is replaced by "from what does quantum mechanics emerge?" — and that question has a definite answer: from the topology, admissibility geometry, and projection structure of the phase substrate.

Quantum mechanics is not wrong. Its predictions are correct, its mathematical structure is beautiful, and its empirical scope is extraordinary. But it is secondary. It is the description-level shadow of a deeper reality — a reality that is not made of particles, fields, spacetime, or probability amplitudes, but of phase.

Reality is phase.

Appendix A: Complete Proof of the Born Rule Theorem (Theorem 25.1)

We provide the full measure-theoretic proof of Theorem 25.1. The proof requires the following lemma.

Lemma A.1. In the linearization regime, the phase density $\rho_\psi(c) = |\psi(\Pi(c))|^2$ is a probability density on $(C_{adm}, \mu_{overlap})$ satisfying $\int_{C_{adm}} \rho_\psi \, d\mu_{overlap} = 1$.

Proof of Lemma A.1. By Proposition 9.1, $\psi \in Q$ is a square-integrable function on Q. The pull-back $\Pi^*\psi$ of ψ to C_{adm} via Π defines the phase density $\rho_\psi = |\Pi^*\psi|^2$. By the change-of-variables formula for the push-forward measure $\Pi_*(\mu_{overlap})$ on Q: $\int_{C_{adm}} |\Pi^*\psi|^2 \, d\mu_{overlap} = \int_Q |\psi|^2 \, d(\Pi_*\mu_{overlap}) = \|\psi\|^2_{Q} = 1$. □

Proof of Theorem 25.1.

Setup. Let $\{|\alpha_n\rangle\}_{n=1}^N$ be the eigenstates of the observable $\hat{O}$ in $\mathscr{H}$ (we treat the discrete case; the continuous case follows by replacing sums with integrals). Let $\psi \in \mathscr{H}$ be normalized: $\|\psi\| = 1$. We write $\psi = \Sigma_n c_n |\alpha_n\rangle$ where $c_n = \langle\alpha_n|\psi\rangle$.

Step 1: Phase density decomposition. The preimage $\Pi^{-1}(\psi)$ decomposes over the basin partition: $\Pi^{-1}(\psi) = \cup_n [\Pi^{-1}(\psi) \cap B_n] \cup R$, where B_n is the admissibility basin corresponding to eigenstate $|\alpha_n\rangle$ and R is a set of measure zero (the basin boundaries). By Lemma A.1: $\Sigma_n \mu_{overlap}(\Pi^{-1}(\psi) \cap B_n) = \int_{C_{adm}} \rho_\psi \, d\mu_{overlap} = 1$.

Step 2: Computation of intersection measure. In the linearization regime, the phase density ρ_ψ restricted to the preimage of the n-th eigenspace is: $\rho_\psi|_{\Pi^{-1}(|\alpha_n\rangle)} = |c_n|^2 \cdot \rho_{\alpha_n}$, where ρ_{α_n} is the unit-normalized phase density of the eigenstate $|\alpha_n\rangle$. Therefore: $\mu_{overlap}(\Pi^{-1}(\psi) \cap B_n) = \int_{B_n} \rho_\psi \, d\mu_{overlap} = |c_n|^2 \cdot \int_{B_n} \rho_{\alpha_n} \, d\mu_{overlap} = |c_n|^2$.

Step 3: Born Rule. By Definition 25.1: $P(\alpha_n|\psi) = \mu_{overlap}(B_n \cap \Pi^{-1}(\psi)) / \mu_{overlap}(\Pi^{-1}(\psi)) = |c_n|^2 / 1 = |\langle\alpha_n|\psi\rangle|^2$. □

Completeness check: $\Sigma_n P(\alpha_n|\psi) = \Sigma_n |c_n|^2 = \|\psi\|^2 = 1$. ✓

Continuous spectrum case: The same argument applies with the Lebesgue measure on the spectrum of Ô replacing the discrete sum: $P(d\lambda|\psi) = |\langle dE_\lambda|\psi\rangle|^2 = d\|E_\lambda \psi\|^2$, where E_λ is the spectral measure of Ô. This reproduces the continuous Born rule in standard QM. □

Appendix B: Comparison of QM Postulates and PPQT Theorems

QM Postulate	Content	PPQT Theorem/Proposition	Section
1. State Space	State space of a system is a Hilbert space $\mathscr{H}$ over $\mathbb{C}$	Theorem 10.1 (Hilbert Space Emergence)	§10
2. Observables	Physical observables correspond to self-adjoint operators on $\mathscr{H}$	Proposition 14.1 (Hermiticity from Admissibility)	§14
3. Measurement Outcomes	Measurement of observable Ô yields an eigenvalue λ_n	Theorem 15.1 (Spectral Derivation Theorem)	§15
4. Born Rule	$P(\lambda_n|\psi) = |\langle\lambda_n|\psi\rangle|^2$	Theorem 25.1 (Born Rule Theorem)	§25
5. State Collapse	After measurement of λ_n, state updates to $|\lambda_n\rangle$	Theorem 22.1 (Measurement Theorem) — collapse replaced by admissibility	§22

QM Postulate	Content	PPQT Theorem/Proposition	Section
		pruning	
6. Schrödinger Evolution	Between measurements: $i\hbar\, \partial\psi/\partial t = \hat{H}\psi$	Theorem 26.1 (Schrödinger Emergence Theorem)	§26
7. Composite Systems	State space of composite system is $\mathscr{H}_A \otimes \mathscr{H}_B$	Theorem 19.1 + Proposition 20.1	§19, §20

Appendix C: Glossary of PPQT Terms

Admissibility Basin (B_α): A connected component of C_{adm}; a maximal set of admissible phase configurations mutually reachable by continuous deformation. §4.3

Admissibility Functional (Φ): The pruning functional $\Phi: C \to \{0,1\}$ determining which phase configurations are physically realized. §5.1

Admissibility Sharpening: Progressive deepening of effective basin depth due to environmental coupling, leading to decoherence. §24

Basin Depth (δ_α): The characteristic depth of admissibility basin B_α, measuring the energy barrier to making a configuration inadmissible. §5.2

Born Rule: The theorem $P(\alpha|\psi) = |\langle\alpha|\psi\rangle|^2$, derived as a measure-theoretic result about phase volume ratios. §25

Classical Limit: The regime in which all basins are deep and well-separated, yielding $O(\alpha,\beta) \approx 0$. §35

Critical Depth (δ_crit): The threshold basin depth separating shallow (quantum) from deep (classical) basins. §5.2

Decoherence: Admissibility sharpening induced by environmental coupling; suppression of quantum interference. §24

Empirical Window: The set of physical regimes accessible to experimental measurement at a given epoch. §34

Entanglement: The condition C_adm^{AB} ≠ C_adm^A × C_adm^B — non-factorizability of the composite admissible configuration space. §19

Global Consistency Functional (GC): GC: C → ℝ≥0, measuring degree of topological incoherence; GC(c) = 0 iff c ∈ C_adm. §4.1

Hilbert Space ($\mathcal{H}$): The Cauchy completion of the reduced descriptive space Q in the linearization regime. §10

Inner Product: ⟨φ|ψ⟩ = ∫_{C_adm} φ*(c) ψ(c) dμ_overlap(c) — phase overlap integral. §10

Linearization Regime: The regime of small phase variations, shallow basins, and locally trivial topology. §9

Macroscopic Phase Structure: A phase structure M with deep basin (δ_M >> δ_crit) and topological isolation. §23

Measurement: A phase coupling that prunes system admissibility basins to a single basin; no collapse. §22

Overlap Measure (O(α,β)): The normalized intersection measure of two basin projections: O(α,β) = μ_overlap(Π(B_α) ∩ Π(B_β)) / μ_overlap(Π(B_α) ∪ Π(B_β)). §7

Phase Action Quantum (ℏ): The minimum nonzero admissible phase variation over a closed loop in C_adm. §18

Phase Configuration Space (C): The smooth Fréchet manifold of all possible phase configurations. §4.1

Phase Mode: An independent coherent excitation of C_adm, labeled by topological quantum numbers. §31

Phase Saturation: The regime where local phase density reaches its maximum admissible value. §37

Pruning: See Admissibility Functional, Measurement.

Quantum Regime: The regime of partially overlapping admissibility basins: $0 < O(\alpha,\beta) < 1$. §7

Reduced Description Map (Π): The canonical surjective map Π: C_adm → Q preserving experimentally distinguishable structure. §8

Reduced Descriptive Space (Q): The quotient of C_adm under descriptive equivalence; the target of Π. §8

Separability (PPQT): C_adm^{AB} ≅ C_adm^A × C_adm^B — factorizability of the composite admissible space. §20

Shallow Basin: A basin with δ_α < δ_crit; characteristic of quantum systems. §5.2

Superposition: The condition of multiple admissible basins overlapping under available description, without ontological commitment to simultaneous actuality. §13

Topological Admissibility Class: An equivalence class of configurations under continuous phase deformation within C_adm. §6

Topological Dominant Regime: The regime where topological invariants dominate over metric-geometric features. §36

Wavefunction: A coordinate representation of phase-configuration overlap density on Q; not a physical field. §11

Appendix D: Notation Reference

Symbol	Meaning	Domain / Range	Section
C	Full phase configuration space	Fréchet manifold	§4.1
C_{adm}	Admissible subspace $\{c : GC(c) = 0\}$	Stratified submanifold of C	§4.1
C_{adm}^{AB}	Composite admissible space	Subspace of $C_A \times C_B$	§19
Φ	Admissibility (pruning) functional	$C \to \{0,1\}$	§5.1
GC	Global consistency functional	$C \to \mathbb{R}_{\geq 0}$	§4.1
B_α	Admissibility basin labeled α	Open connected subset of C_{adm}	§4.3
δ_α	Characteristic depth of basin B_α	$\mathbb{R}_{\geq 0}$	§5.2
δ_{crit}	Critical basin depth	$\mathbb{R}_{>0}$	§5.2
ω_α	Width of basin B_α	$\mathbb{R}_{\geq 0}$	§5.2
$O(\alpha,\beta)$	Overlap measure	$[0,1]$	§7

Symbol	Meaning	Domain / Range	Section
	of basins B_α, B_β		
μ_overlap	Phase overlap Borel measure on C_adm	Measure on C_adm	§4.4
d_D	Distinguishability metric	C_adm × C_adm → [0,1]	§4.4
Π	Canonical reduced description map	C_adm → Q	§8
Π*	Pull-back of Π	Functions on Q → functions on C_adm	§A
Q	Reduced descriptive space	Quotient of C_adm	§8
ℋ	Hilbert space (emergent)	Separable complex Hilbert space	§10
ℱ	Fock space of phase modes	⊕_{n=0}^∞ ℋ^⊗_s n	§31
ψ	Wavefunction / state vector	Q → ℂ, or element of ℋ	§11
ρ	Density matrix (mixed state)	Positive trace-1 operator on ℋ	§20
ρ_ψ	Phase density of ψ	L^2(C_adm, μ_overlap)	§A
⟨φ|ψ⟩	Inner product on ℋ	ℂ	§10
‖ψ‖	Norm: √⟨ψ|ψ⟩	ℝ≥0	§10
Ô, Â, B̂, Ĥ	Operators on ℋ	Linear maps ℋ → ℋ	§14
[Â,B̂]	Commutator: ÂB̂ − B̂Â	Linear map on ℋ	§16
G_Φ	Phase symmetry group	Group of admissibility-preserving maps	§29

Symbol	Meaning	Domain / Range	Section
π_0(C_adm)	Connected components of C_adm (topological classes)	Discrete set ≅ ℤ	§6
π_1(C_adm)	Fundamental group of C_adm (loops)	Group	§18
ℏ	Phase action quantum (reduced Planck constant)	ℝ>0 (J·s)	§18
c	Phase propagation speed (speed of light)	ℝ>0 (m/s)	§42
A_μ	Gauge connection one-form	Differential form on B	§30
F_μν	Gauge field strength tensor (curvature)	Two-form on B	§30
â_n, â†_n	Annihilation/ creation operators for phase mode n	Operators on ℱ	§31
γ^μ	Dirac gamma matrices	4×4 complex matrices	§33
S_phase	Phase action functional	Functional on paths in C_adm	§28
κ(P_A, P_B)	Topological resolution bound	ℝ≥0	§17
τ_D	Decoherence timescale	ℝ>0 (s)	§24
Δ_top	Topological gap	ℝ≥0 (J)	§36
ρ_max	Maximum admissible phase density	ℝ>0	§37

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